LM01 Portfolio Management: An Overview

Version 1.0

1. Introduction

In this reading, we will see the importance of the portfolio approach to investing, investment needs of different types of investors, steps in the portfolio management process, and how to compare various types of pooled investments.

2. Portfolio Perspective: Diversification and Risk Reduction

The portfolio approach means evaluating individual investments based on their contribution to the investment characteristics of the portfolio. Assume an investor's portfolio has three stocks A, B, and C. He is evaluating whether to add another stock, D, to the portfolio. In a portfolio approach, the investor will analyze what will happen to the risk and return of the portfolio with and without stock D; whereas, in an isolated approach, he will only look at the merits and demerits of the stock D.

Diversification helps investors avoid disastrous outcomes. For instance, many Enron employees held all of their retirement funds in Enron shares. When the share tumbled from $90 to zero between January 2001 and 2002, it completely ruined their financial wealth; this example emphasizes the need to diversify one's portfolio. Instead, if the Enron employees had held shares of other companies or other products, the consequence would not have been as bad.

Diversification also helps investors reduce risk without compromising their expected rate of return. A simple measure of diversification risk is the diversification ratio. Let us take the example of the same portfolio consisting of three stocks: A, B, and C with each stock belonging to a different industry. There is a high probability that the movements of A, B, and C are not correlated with each other, i.e., when A moves up, B may move down or C may move down. The benefit of diversification is that the movements of individual stocks cancel each other out to some extent.

The benefits of diversification in risk reduction for an equally weighted portfolio is measured by diversification ratio.

$$\text{Diversification ratio} = \frac{\text{Risk of equally weighted portfolio of n securities}}{\text{Risk of single security selected at random}}$$

3. Portfolio Perspective: Risk-Return Trade off, Downside Protection, Modern Portfolio Theory

The composition of the portfolios also matters for the risk-return trade-off. The table below shows two portfolios with different compositions of A, B, and C.

Two portfolios with same stocks but different compositions					
	Weight of each stock				
Portfolio	**Stock A**	**Stock B**	**Stock C**	**Expected Return**	**Standard Deviation**
Portfolio 1	33%	33%	33%	11%	13.1%

Portfolio 2	50%	25%	25%	11%	14%

As you can see both the portfolios have the same expected return, but Portfolio 1 has a better risk-return trade-off than Portfolio 2 as the risk assumed is lower for the same return.

However, an important point to note is that portfolios do not provide guaranteed downside protection. Although portfolio diversification reduces risk, the level of risk reduction is not the same at times of financial crises. The benefits of risk reduction from diversification are best seen under normal market conditions.

4. Steps in the Portfolio Management Process

The three steps in the portfolio management process are planning, execution, and feedback.

Step One: The Planning Step
In this step, the portfolio manager needs to understand a client's needs and develop an investment policy statement (IPS). IPS is a written document that states the client's objectives and constraints. Objectives are return and risk objectives which may be stated in absolute terms or relative terms. Constraints may include liquidity, unique circumstances, time horizon, legal, and taxes.

Step Two: The Execution Step
Based on the IPS, a portfolio is constructed in this step. Under execution, the first activity is asset allocation. Here the portfolio manager decides what asset classes must be included in the client's portfolio and in what proportion. Stocks, bonds, and alternative investments are examples of different asset classes. As an example, a client's asset allocation may consist of 60% equities, 30% fixed-income securities and 10% alternative investments.

Asset allocation is followed by security selection which is the analysis and selection of individual securities. In other words, the specific securities to be purchased are identified. For example, if 60% is allocated to equities, the analyst will identify what specific stocks to purchase.

Once the securities are selected, they are purchased through a broker or dealer. This is called portfolio construction.

Step Three: The Feedback Step
A portfolio manager's responsibility does not end with constructing a portfolio. The portfolio also needs to be monitored and rebalanced at regular intervals. For example, if the stock market performs very well in a particular period, the asset allocation drifts away from the intended levels. In this case the portfolio needs to be rebalanced. The frequency at which performance is measured is pre-decided – for instance, it could be on a monthly or quarterly basis.

The feedback step also involves performance measurement and reporting. Analysis must include how the portfolio performed over time, were the objectives met, what assets attributed to the good/poor performance, how the portfolio performed against the benchmark, and so on.

5. Types of Investors

The investment needs of different client types are given in the following table:

	Investment Horizon	**Risk Tolerance**	**Income Needs**	**Liquidity Needs**
Individual Investors	Depends on individual goals.	Depends on the ability and willingness to take risk.	Depends on rationale behind investment.	Depends on the individual.
Banks	Short	Low	Pay interest on deposits.	High, to meet the daily withdrawals.
DB pension plans	Long, depends on the employee profile.	High for longer investment horizon.	High for mature funds (payouts are closer), low for growing funds.	Low
Endowments and foundations	Long	High	Meet spending obligations.	Low
Insurance Companies (P&C)	Short	Low	Low	High
Insurance Companies (Life)	Long	Low (because of high liquidity needs).	Low	High
Mutual Funds	Varies by fund.	Varies by fund.	Varies by fund.	High, to meet redemptions.

Instructor's Note:

The two types of pension plans are:

Defined Contribution Plan: Company contributes an agreed-upon amount to the plan. The agreed-upon amount is recognized as a pension expense on the income statement and the contributed amount is treated as an operating cash outflow. In DC plan, the investment and inflation risk is borne by the employee.

Defined Benefit Plan: A company makes promises of future benefits to be paid to the employees. Since the employee's future benefits are defined, the employer assumes the investment risk.

6. The Asset Management Industry

The asset management industry is an integral component of the global financial services sector. At the end of 2017, the industry managed more than US$79 trillion of assets owned by a broad range of institutional and individual investors.

Asset managers are usually referred to as a buy-side firm since it uses (buys) the services of sell-side firms. A sell-side firm is a broker/dealer that sells securities and provides independent investment research and recommendations to buy-side firms.

Asset managers offer a wide variety of strategies (e.g., emerging market equities or quantitative investing). Some asset managers are "multi-boutique," in which a holding company owns several asset management firms that offer specialized investment strategies.

Active versus Passive Management
Asset managers can provide active or passive management or both.

- Active asset managers attempt to outperform a pre-determined performance benchmark.
- Passive asset managers attempt to match returns of a pre-determined benchmark.

Traditional versus Alternative Asset Managers
Asset managers are typically categorized as either traditional or alternative.

- Traditional asset managers focus on long-only equity, fixed-income, and multi-asset investment strategies. In traditional asset management, the management fee is based on asset under management.
- Alternative asset managers focus on hedge fund, private equity, and venture capital strategies, etc. Alternative asset managers' revenue is based on management fee and performance fees (or "carried interest").

Since many traditional managers are now offering higher-margin alternative products to clients, the line between traditional and alternative managers is becoming blurred.

Ownership Structure
The majority of asset management firms are privately owned and are structured as limited liability companies or limited partnerships. The publicly traded asset managers are not very common but they have substantial assets under management. Another prevalent ownership form of asset management is asset management divisions of large, diversified financial services companies that offer asset management alongside insurance and banking services.

Asset Management Industry Trends
The three key trends in the asset management industry include the following:

1) Growth of passive investing due to low cost for investors and difficulty for active managers to generate ex ante alpha in more-efficiently priced markets.

2) "Big data" in the investment process: Big data refers to massively large datasets and their analysis. These data include

- Structured data i.e. order book data and security returns.
- Unstructured data i.e. data generated by a vast number of activities on the internet and elsewhere (e.g., compiled search information).

Advanced statistical and machine-learning techniques are being used by asset managers to help process and analyze these new sources of data. Third-party research vendors are supplying new sources of data, including social media data (i.e. data provided by Twitter and Facebook) and imagery and sensor data (e.g., weather conditions, cargo ship traffic patterns and company-specific considerations like retailer parking capacity/usage, tracking of retail customers).

3) Robo-advisers in the wealth management industry: It refers to use of automation and investment algorithms to provide several wealth management services, such as investment planning, asset allocation, tax loss harvesting, and investment strategy selection. They key drivers that are leading to rapid growth in robo-advisory assets include growing demand from "mass affluent" and young investors, lower fees, and new entrants' opportunities owing to low barriers to entry.

7. Pooled Interest - Mutual Funds

Pooled investments are where money is collected from several individual investors to be invested in a large portfolio. As the name implies, it is pooling money together for an investment. The funds where this collected money is invested could range from mutual funds to private equity depending on the risk, capital required, strategy, and how it is managed. The different types of investment products generally available to investors are listed below:

Investment Products by Minimum Investment	
Investment Product	**Minimum Investment**
Mutual Funds	$50 +
Exchange Traded Funds (ETFs)	$50 +
Separately Managed Accounts	$100,000 +
Hedge Funds	$1,000,000 +
Private Equity Funds	$1,000,000 +

In the rest of this section, we understand what a mutual fund is, the different types of mutual funds, and how other investment products are different from mutual funds.

7.1 Mutual Funds

A mutual fund is a comingled investment pool in which each investor has a pro-rata claim on the income and value of the fund.

Consider the following example: An investment firms raises $100,000 for a stock-based mutual fund from five investors and issues 10,000 shares. Each share has a value of $10.

There are three individual investors and two institutional investors. The number of shares is based on the amount invested relative to the total amount.

Investor	Amount Invested	% of Total	Number of Shares
John	$4,000	4%	400
Jill	$6,000	6%	600
Joe	$10,000	10%	1,000
Jones Co.	$50,000	50%	5,000
Widget Co.	$30,000	30%	3,000

Assume the $100,000 is invested in various stocks and it grows to $150,000. The value of each share goes up by 50% to $15. The advantage of this structure is that the investment firm can have one or two managers managing this entire pool of money and each individual investor need not hire a manager to manage his relatively small amount of money. This is a cost-effective way of managing money.

Advantages of Mutual funds:

- Low investment minimums
- Diversified portfolios
- Daily liquidity
- Standardized performance and tax reporting.

In the context of mutual funds, it is important to understand the following terms:

- <u>Net asset value</u>**:** Net asset value = value of assets – liabilities. The value of a mutual fund is called the net asset value. It is calculated on a daily basis based on the closing price of the stocks held in the fund's portfolio. The NAV per share is calculated as: NAV/number of total shares. The NAV per share in our previous example was $100,000/10,000 = $10 per share
- <u>Open-end fund</u>**:** A mutual fund with no restrictions on when new shares can be issued or when funds can be withdrawn. The fund accepts new investment money and issues additional shares at a price equal to the net asset value at the time of investment. Similarly, when an investor redeems shares, the fund sells the underlying assets/securities to retire so many shares at the current net asset value. Because of this, an open-end fund trades close to NAV. NAV is based on closing prices. They can be bought/sold only once during the day. They are also called evergreen funds.
- <u>Closed-end fund</u>**:** Unlike open-end fund, in a closed-end fund, no new investment money is accepted. Shares of closed-end funds trade in the secondary market. A new investor may invest in the fund if an existing investor is willing to sell his shares. So, the outstanding shares stay the same. Since there is no liquidation of underlying assets and the share base is unchanged, the NAV may trade either at a premium or discount to net asset value based on the demand for shares. The units issued by closed-end funds trade like regular shares – they can be bought or sold on margin, shorted, etc.

Mutual funds can also be classified into:

- No-load fund**:** Most mutual funds have an annual fee for managing the fund, which is a percentage of the fund's net asset value. In a no-load fund, only an annual fee is charged, but there is no fee for investment or redemption.
- Load fund: A percentage is charged for investment or redemption or both (called entry and exit load) in addition to the annual fee.

8. Pooled Interest - Types of Mutual Funds

Funds can be categorized based on types of assets they invest in:

- Money market – taxable or non-taxable: They invest in high quality, short-term debt. Taxable money market bonds invest in corporate debt and federal government debt, while tax-free bonds invest in state and local government debt.
- Bond mutual funds – taxable or non-taxable: They invest in a portfolio of individual bonds and preference shares.
- Stock/index mutual funds – domestic or international: They invest in a portfolio of stocks or index funds.
- Hybrid/balanced funds – They invest in both stocks and bonds.

Funds can be categorized as actively managed or passively managed:

- With actively managed funds, the manager tries to identify securities which will outperform the market; these funds have high fees relative to passively managed funds.
- With passively managed funds, the manager purchases the same securities as a benchmark index. This helps ensure that the performance of the fund is similar to the performance of the benchmark.

9. Pooled Interest – Other Investment Products

Separately Managed Accounts

- A separately managed account (SMA) is an investment portfolio managed privately for an individual or institution by a brokerage firm or individual investment professional (financial advisor).
- Also called managed account, wrap account, and individually managed account.
- SMAs are managed exclusively for the benefit of a single individual or institution to meet their needs with respect to objectives, risk tolerance, and constraints.
- Unlike a mutual fund, the assets of an SMA are owned directly by the individual or institution.
- The required minimum investment in SMA is much higher than that for a mutual fund.

9.1 Exchange Traded Funds

Like mutual funds, ETFs are a pooled investment vehicle often based on an index. With index mutual funds, investors buy shares directly from the fund. With ETFs, investors buy shares from other investors.

How ETFs work

A fund manager creates the ETF by determining what assets the ETF will hold. Once the securities are decided, the fund sponsor contacts an institutional investor who owns those securities. The institutional investor deposits the basket of securities with the fund sponsor (held through a custodian), and in return, receives creation units for the deposited securities. The creation units typically represent 50,000 to100,000 ETF shares.

It is important to note that the weight of securities deposited is often in the proportion of what it is trying to represent. For example, the weight of Athena Health in iShares Russell 2000® Growth Index Fund is 0.61%, and it is roughly the same as in Russell 2000® Growth Index. The institutional investor then sells the creation units as ETF shares to the public; investors buy shares from other investors, so it works like a closed mutual fund. The institutional investor may redeem the original securities by returning the ETF creation units.

How ETFs are Similar to Mutual Funds

ETFs combine features of closed-end and open-end funds:

- Trade like closed-end mutual funds; can be shorted and bought on margin.
- Because of unique redemption procedure, their prices are close to net asset value like open-end funds.
- Expenses tend to be low relative to mutual funds but brokerage fee needs to be paid.
- Unlike mutual funds, ETFs do not have capital gains distributions and the minimum required investment in ETFs is usually smaller than that of mutual funds.

9.2 Hedge Funds

Hedge funds are private investment vehicles that typically use leverage, derivatives, and long and short investment strategies.

Characteristics of hedge funds:

a) *Short selling*: Many hedge funds use short-selling directly or synthetically using options, futures, and credit default swaps.

b) *Absolute return seeking:* Hedge funds seek an absolute, positive returns in all market environments.

c) *Leverage:* Most of the hedge funds use financial leverage (bank borrowing) or implicit leverage (using derivatives).

d) *Low correlation:* Some hedge funds tend to have low return correlations with traditional equity and/or fixed-income asset classes.

e) *Fee structures:* Hedge funds typically charge two distinct fees: a traditional asset-based management fee (AUM fee) and an incentive (or performance) fee.

9.3 Private Equity and Venture Capital Funds

Private equity and venture capital funds are privately held and actively managed equity positions. In such funds, a firm makes an investment in a company and then is actively involved in the management of that company. The equity they hold is private and not traded in public markets. The intention is to exit out of the investment in a few years. Like majority of alternative funds, private equity and venture capital funds are structured as limited partnerships between the fund manager, called the general partner (GP), and the fund's investors, called limited partners (LPs). The revenue of these funds comprises of:

- management fee (fees based on committed capital or net asset value or invested capital)
- Transaction fees
- *Carried interest:* GP's share of profits (typically 20%)
- *Investment income.* Profits generated on capital contributed to the fund by the GP.

Summary

LO: Describe the portfolio approach to investing.

The portfolio approach means evaluating individual investments by their contribution to the risk and return of an investor's portfolio.

Diversification helps investors reduce risk without compromising their expected rate of return. A simple measure of diversification risk is the diversification ratio. The lower the ratio, the better the diversification.

$$\text{Diversification ratio} = \frac{\text{Risk of equally weighted portfolio of n securities}}{\text{Risk of single security selected at random}}$$

LO: Describe the steps in the portfolio management process.

The three steps in the portfolio management process are:

1. Planning	• Understanding the client's needs. • Preparing an Investment policy statement.
2. Execution	• Asset allocation. • Security analysis. • Portfolio construction.
3. Feedback	• Portfolio monitoring and rebalancing. • Performance measurement and reporting.

LO: Describe types of investors and distinctive characteristics and needs of each.

The investment needs of each client type are given in the following table:

	Investment Horizon	**Risk Tolerance**	**Income Needs**	**Liquidity Needs**
Individual Investors	Depends on individual goals	Depends on the ability and willingness to take risk	Depends on rationale behind investment	Depends on individual
Banks	Short	Low	Pay interest on deposits	High, to meet the daily withdrawals
DB pension plans	Long, depends on the employee profile	High for longer investment horizon	High for mature funds (payouts are closer), low for growing funds	Low
Endowments and foundations	Long	High	Meeting spending obligations	Low

Insurance Companies (P&C)	Short	Low	Low	High
Insurance Companies (Life)	Long	Low (because of high liquidity needs)	Low	High
Mutual Funds	Varies by fund	Varies by fund	Varies by fund	High, to meet redemptions

LO: Describe defined contribution and defined benefit pension plans.

Defined Contribution Plan: Company contributes an agreed-upon amount to the plan. The agreed-upon amount is recognized as a pension expense on the income statement and the contributed amount is treated as an operating cash outflow.

Defined Benefit Plan: A company makes promises of future benefits to be paid to the employees.

LO: Describe aspects of the asset management industry.

An asset management industry is an integral component of the global financial services sector.

Asset managers are usually referred to as a buy-side firm as it uses (buys) the services of sell-side firms (broker/dealer that sells securities and provides independent investment research and recommendations to buy-side firms).

- Active asset managers attempt to outperform pre-determined performance benchmark.
- Passive asset managers attempt to match returns of a pre-determined benchmark.
- Traditional asset managers focus on long-only equity, fixed-income, and multi-asset investment strategies.
- Alternative asset managers focus on hedge fund, private equity, and venture capital strategies, etc.
- The majority of asset management firms are privately owned and are structured as limited liability companies or limited partnerships.
- The three key trends in the asset management industry include growth of passive investing, "Big data" in the investment process, and robo-advisers in the wealth management industry,

LO: Describe mutual funds and compare them with other pooled investment products.

A mutual fund is a comingled investment pool in which each investor has a pro-rata claim on the income and value of the fund. In the context of mutual funds, it is important to understand the following terms:

Net asset value: Net asset value = value of assets – liabilities.

The value of a mutual fund is called the net asset value.

Open-end fund: A mutual fund with no restrictions on when new shares can be issued or when funds can be withdrawn.

Closed-end fund: Unlike an open-end fund, in a closed-end fund no new investment money is accepted.

No-load fund: Only an annual fee is charged, but there is no fee for investment or redemption.

Load fund: A percentage is charged for investment or redemption or both, in addition to the annual fee.

Funds can be categorized based on the types of investments (money market, bond mutual funds, stock mutual funds, index funds).

Like mutual funds, exchange traded funds (ETFs) are a pooled investment vehicle, often based on an index. With index mutual funds, investors buy shares directly from the fund. With ETFs, investors buy shares from other investors.

ETFs combine features of closed-end and open-end funds.

- Track NAV like open-end funds.
- Trade like close-end funds (continuously traded, can buy on margin, can short sell).
- Expenses are lower compared to mutual funds, but brokerage fee needs to be paid.
- Unlike mutual funds, ETFs do not have capital gains distributions.

Separately Managed Accounts (SMA)

- Also called “managed account”, “wrap account”, “individually managed account.”
- The investor owns individual (underlying) shares.
- Tax implications are considered when buying or selling.
- Requires high minimum investment.

Hedge Funds

- Hedge funds are private investment vehicles that typically use leverage, derivatives, and long and short investment strategies.

Private Equity and Venture Capital Funds

- Private Equity and Venture Capital Funds are privately held and actively managed equity positions. The equity they hold is private with the intention to exit out of the investment in a few years. Like majority of alternative funds, private equity and venture capital funds are structured as limited partnerships between the fund manager, called the general partner (GP), and the fund’s investors, called limited partners (LPs). The revenue of these funds comprises of management fee, transaction fee, carried interest and investment income.

LM02 Portfolio Risk and Return: Part I

Version 1.0

1. Introduction

This reading covers:

- Investment characteristics of assets in terms of their return and risk.
- How to determine what assets are appropriate for a portfolio.
- How to construct an indifference curve and use it in the selection of an optimal portfolio using two risky assets.
- How to construct an optimal risky portfolio.

Investment Characteristics of Assets: Return (Prerequisite Material)

Return

Return can come in two forms:

- Periodic income through interest payments or dividends.
- Capital gains when the price of the asset you hold increases.

We now look at the various types of return measures and their applicability.

Holding Period Return

Holding period return is the return earned on an asset during the period it was held. It is calculated as a sum of capital gain (price appreciation) and periodic income.

$\text{HPR} = \frac{P_T - P_0 + I_T}{P_0}$

where:

P_T - P_0 = capital gain component

I_T = Income earned during period T

P_T = price at the end of the period

P_0 = price at the beginning of the period

Arithmetic or Mean Return

Arithmetic return is a simple arithmetic average of returns. Assume you have three stocks A, B, and C with returns of 10%, 20%, and 30% respectively. The collective return from the three stocks is (10 + 20 + 30)/3 = 20%.

Geometric Mean Return

Geometric mean return is the compounded rate of return earned on an investment.

$$\text{Geometric mean return} = [(1 + R_{i1}) * (1 + R_{i2}) * \ldots\ldots * (1 + R_{iT})]^{\frac{1}{T}} - 1$$

Assume you have a stock A which returns 10%, 20%, and 30% in years 1, 2, and 3 respectively.

What is the mean return earned?

Geometric mean return = $[(1.1)\ (1.2)\ (1.3)]^{0.333} - 1$ = 19.7%

Money-Weighted Return or IRR (Prerequisite material)

Money-weighted return is the internal rate of return on money invested that considers the cash inflows and cash outflows, and calculates the return on actual investment.

Money-weighted return is a useful performance measure when the investment manager is responsible for the timing of cash flows. This is often the case for private equity fund managers.

Example

Given the data below, compute the holding period return, arithmetic mean return, geometric mean return, and money-weighted return. Assume no withdrawals except at the end of year 3.

Year	Assets under management at start of year (millions of $)	Net return
1	30	10%
2	33	-5%
3	35	15%

Solution:

Holding period return:

Holding period = 3 years

Return = 1.1 x 0.95 x 1.15 = 1.20175 = 20.175%

Arithmetic mean:

Return = (10 – 5 + 15)/3 = 6.67%

Geometric mean:

Return = $(1.1 \text{ x } 0.95 \text{ x } 1.15)^{1/3} - 1 = 6.317\%$

Money-weighted return:

To calculate the money-weighted return, we must know the net cash flows (cash inflows and outflows) for every year. So, let us draw a table and fill in the values and derive some others to get the values for CF_0, CF_1, CF_2, and CF_3.

	Year 1	Year 2	Year 3
Balance from previous year	0	33.00	31.35
New investment by investor	30.00	0	3.65
Withdrawal by investor	0	0	0
Net balance at start of year	30.00	33.00	35.00
Investment return for year	10%	-5%	15%
Investment gain (loss)	3.00	(1.65)	5.25
Balance at end of year	33.00	31.35	40.25

Now that we have the cash flows for the three years, let's use the financial calculator to calculate IRR. $CF_0 = -30$; $CF_1 = 0$; $CF_2 = -3.65$; $CF_3 = +40.25$

IRR = 6.62%

Time-Weighted Rate of Return (Prerequisite material)

The time-weighted rate of return measures the compound growth rate of $1 initially invested in the portfolio over a stated measurement period.

The time-weighted return can be calculated using the following steps:
1. Break the overall evaluation period into sub-periods based on the dates of cash inflows and outflows.
2. Calculate the holding period return on the portfolio for each sub-period.
3. Link or compound holding period returns to obtain an annual rate of return for the year (the time-weighted rate of return for the year).
4. If the investment is for more than a year, take the geometric mean of the annual returns to obtain the time-weighted rate of return over that measurement period.

Consider the following example:

Time	*Outflow*	*Inflow*
0	$20.00 to purchase the first share	
1	$22.50 to purchase the second share	$0.50 dividend received on first share
		$1.00 dividends ($0.50 x 2 shares); $47.00 from selling 2 shares @ $23.50 per share

Calculating the TWRR for this example is relatively simple because cash flows only occur at the start/end of every year. We will follow the steps mentioned earlier:

<u>Steps 1</u>: Break into evaluation period and value the portfolio at start/end of every period.
- Value of the portfolio at the start of Year 1 (t = 0) is $20.00.
- Value of portfolio at the end of Year 1 (t = 1) before the purchase of the new share is 22.50 + 0.50 = $23.00. Note that the dividend of $0.50 on the first share is received at the end of Year 1.
- Value of the portfolio at the start of Year 2 (t = 1) after the purchase of the second share is 22.50 + 22.50 = $45.00. The dividend of $0.50 from the first share is paid out and is not considered as part of the portfolio.
- Value of the portfolio at the end of Year 2 (t = 2) is 23.50 + 23.50 + 0.50 + 0.50 = $48.00. Both shares pay a dividend of $0.50 at the end of the second year.

<u>Step 2</u>: Calculate the holding period return on the portfolio for each sub-period.

- In this question the cash flows are taking place at the start/end of each period. Hence there are no sub-periods. *Scenarios involving sub-periods will be covered in the next example.*

Step 3: Link or compound holding period returns to obtain an annual rate of return for the year.
- The annual rate of return is based on the portfolio value at the start and end of each period.
- The portfolio value at the start of Year 1 was \$20.00 and the value at the end of Year 1 was \$23.00. Hence the holding period return was 15.00%.
- The portfolio value at the start of Year 2 was \$45.00 and the value at the end of Year 2 was \$48.00. Hence the holding period return was 6.67%.

Step 4: If the investment is for more than a year, take the geometric mean of the annual returns to obtain the time-weighted rate of return over that measurement period.

The TWRR is calculated as: $(1.15 \ * \ 1.067)^{\frac{1}{2}} - \ 1 = 0.1077 = 10.77\%$.

Money-weighted v/s time-weighted returns

- The money-weighted rate of return is impacted by the timing and amount of cash flows.
- The time-weighted rate of return is not impacted by the timing and amount of cash flows.
- If funds are added to an investment portfolio just before a period of relatively high returns, the money-weighted rate of return will be higher than the time-weighted rate of return, and vice versa.
- The time-weighted return is an appropriate performance measure if the portfolio manager does not control the timing and amount of investment.
- On the other hand, money-weighted return is an appropriate measure if the portfolio manager has control over the timing and amount of investment.

Annualized Return (Prerequisite material)

Annualized return converts the returns for periods that are shorter or longer than a year, to an annualized number for easy comparison.

$$\text{Annualized return } = \left(1 + r_{\text{period}}\right)^{c} - 1$$
$$\text{Where } c = \text{number of periods in year}$$

Other Major Return Measures (Prerequisite material)

Gross Return

Gross return is the return earned by an asset manager prior to deducting management fees and taxes. It measures the investment skill of a manager.

Net Return

Net return is the return earned by the investor on an investment after all managerial and administrative expenses have been accounted for. This is the measure of return that should matter to an investor.

Assume an investment manager generates $120 for every $100, and charges a 2% fee for management and administrative expenses. The gross return, in this case, is 20% and the net return is 18%.

Pre-tax and After-tax Nominal Return

The returns we saw till now were pre-tax nominal returns, i.e., before deducting any taxes or any adjustments for inflation. This is the default, unless otherwise stated.

After-tax nominal return is the return after accounting for taxes. The actual return an investor earns should consider the tax implications as well.

In the example that we saw above for gross and net return, 18% was the pre-tax nominal return. If the tax rate for the investor is 33.33%, then the after-tax nominal return will be 18(1 - 0.3333) = 12.0006%.

Real Return

Real return is the return after deducting taxes and inflation.

$$(1 + r) = (1 + r_{real})(1 + \pi)$$

where:

r_{real} = real rate

π = rate of inflation

r = nominal rate

In the previous example, the after-tax nominal return was 12%. Assume the inflation rate for the period is 10%. What is the real rate of return?

Using the above formula, $(1 + 0.12) = (1 + r)(1 + 0.1)$. Solving for r, we get 1.818%.

Instructor's tip: If the answer choices are close to each other, use this formula to determine the correct answer. Else, you may use an approximation to solve for r quickly as nominal rate = real rate + inflation.

Leveraged Return

In cases, where an investor borrows money to invest in assets like bonds or real estate, the leveraged return is the return earned by the investor on his money after accounting for interest paid on borrowed money.

Portfolio Expected Return and Variance

For a two-asset portfolio, the expected return and variance can be computed as:

$$E(R_P) = w_1R_1 + w_2R_2$$

$$\sigma_p^2 = w_1^2\,\sigma_1^2 + w_2^2\,\sigma_2^2 + 2\,w_1\,w_2\,\sigma_1\,\sigma_2\,\rho$$

Example

My portfolio consists of two stocks X and Y. X represents 60% of the portfolio and Y the remaining 40%. X has an expected return of 12% and a standard deviation of 16%. Y has an expected return of 20% and a standard deviation of 30%. The correlation is 0.5. What is the expected return and risk of my portfolio? How does the return/risk change when the weights of X and Y change?

Solution:

$E(R_P) = w_1R_1 + w_2R_2$

$E(R_P) = 0.6 \text{ x } 12 + 0.4 \text{ x } 20 = 15.2\%$

We can calculate the portfolio variance:

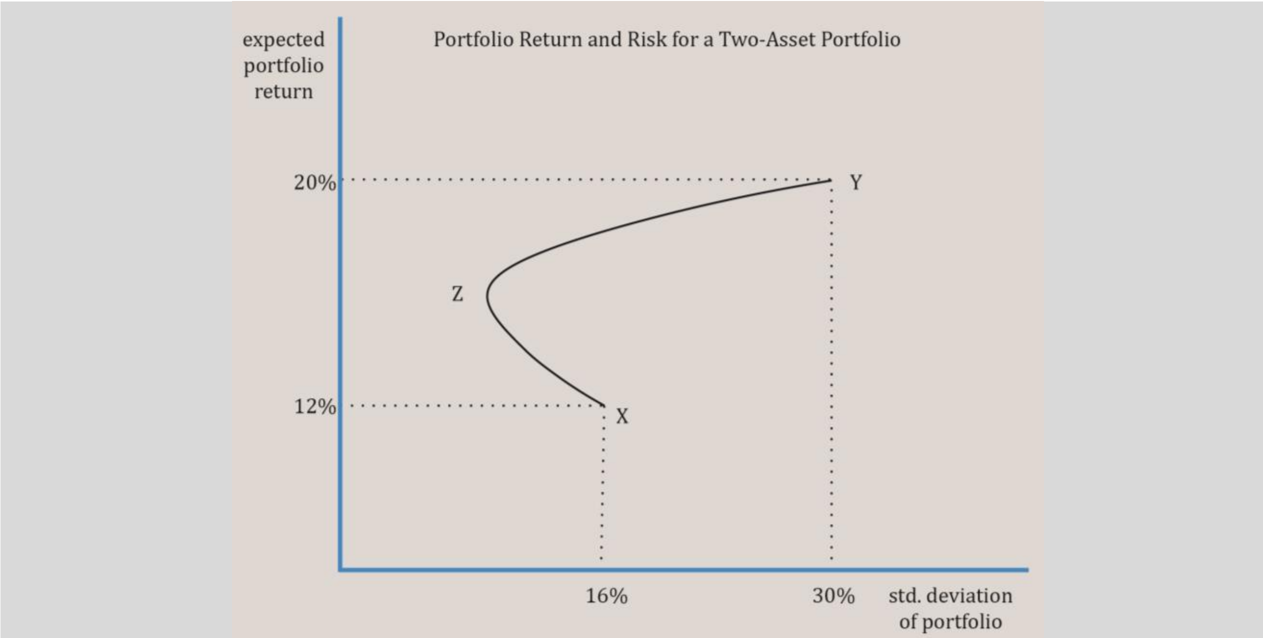
$\sigma_p^2 = (0.6)^2 (0.16)^2 + (0.4)^2 (0.3)^2 + 2 (0.6) (0.4) (0.16) (0.3) (0.5) = 0.0351$

$\sigma_p = 0.1874 = 18.74\%$

To understand how return/risk change when the weights of X and Y change, we will calculate the risk and return for different weights and plot a curve as shown in the graph below.

- Point X represents 100% of the portfolio invested in stock X with return of 12% and standard deviation of 16%.
- Point Y represents 100% of the portfolio invested in stock Y with expected return of 20% and a standard deviation of 30%.
- The curve between X and Z (to the left of X) represents the region where amount invested in stock X is decreased while the weight of Y is increased. This region is where the benefit of diversification is seen, i.e., the expected return increases while risk goes lower.
- Beyond the Point Z, when the weight of Y is increased till the point Y where 100% is invested in Y, there is no diversification benefit. For any point between Z and Y, the risk and return increase.

The graph below plots portfolio risk and return for a two-asset portfolio and shows the impact of correlation of assets on portfolio risk. As you can see, there is no risk-return trade-off when 100% is invested either in X or Y.



2. Historical Return and Risk

Historical return is the return actually earned in the past, while expected return is the return one expects to earn in the future.

Historical data shows that higher returns were earned in the past by assets with higher risk. Of the three major asset classes in the U.S., namely stocks, bonds, and T-bills, it has been observed that stocks had the highest risk and return followed by bonds, and T-bills had the lowest return and lowest risk.

Asset Class	Annual Average Return	Standard Deviation (Risk)
Small-cap stocks	High ↓ Low	High ↓ Low
Large-cap stocks		
Long-term corporate bonds		
Long-term treasury bonds		
Treasury bills		

3. Other Investment Characteristics

In evaluating investments using mean and variance, we make the following two assumptions:

- Assumption 1: Distributional characteristics.
  Returns are normally distributed. If this assumption does not hold, then consider skewness and kurtosis. These concepts have been covered earlier in quantitative methods.

- Assumption 2: Market characteristics.
  Markets are informationally and operationally efficient. If markets are not operationally efficient, then it leads to: 1) high transaction costs 2) wide bid-ask spreads and 3) larger price impact of trades.

4. Risk Aversion and Portfolio Selection & The Concept of Risk Aversion

The Concept of Risk Aversion

Risk aversion refers to the behavior of investors preferring less risk to more risk.

Risk tolerance is the amount of risk an investor is willing to take for an investment. High risk aversion is the same as low risk tolerance. Three types of risk profiles are outlined below:

- **Risk Seeking**: A risk seeking investor prefers high return and high risk. Given two investments, A and B, where both have the same return but investment A has higher risk, he will prefer investment A.
- **Risk Neutral**: A risk neutral investor is concerned only with returns, and is indifferent about the risk involved.
- **Risk Averse**: A risk averse investor will prefer an investment that has lower risk, all else equal. Given two investments, A and B, where both have the same return but investment A has higher risk, he will prefer investment B.

5. Utility Theory and Indifference Curves

The utility of an investment can be calculated as:

$$\text{Utility} = E(r) - 0.5 \times A \times \sigma^2$$

where:
A = measure of risk aversion (the marginal benefit expected by the investor in return for taking additional risk. *A* is higher for risk-averse individuals.
σ^2 = variance of the investment
E(r) = expected return

Instructor's Note:
While using this formula, use only decimal values for all parameters.

Example

An investor with A = 2 owns a risk-free asset returning 5%. What is his utility?

Solution:

Utility = 0.05 - 0.5 x 2 x 0 = 0.05

Now, he is considering an asset with σ = 10%. At what level of return will he have the same utility?

0.05 = E(r) – 0.5 x 2 x 0.1^2. Solving for E(R) we get 0.06 or 6.00%.

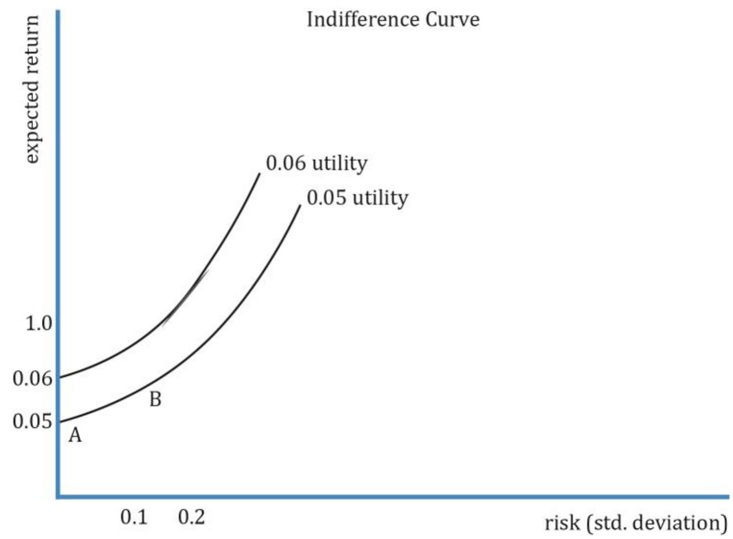
Given a choice between a risk-free asset and stock with an expected return of 10% and σ = 20%, what will he prefer?

U = 0.1 - 0.5 * 2 * 0.2^2 = 0.06. Since the utility number of 0.06 is higher than 0.05, the investor will prefer this investment over the previous one.

Indifference Curves

The indifference curve is a graphical representation of the utility of an investment. An indifference curve plots various combinations of risk-return pairs that an investor would accept to maintain a given level of utility. If the combinations of risk-return on a curve provide the same level of utility, then the investor would be indifferent to choosing one. Each point on an indifference curve shows that the investor is indifferent to what investment he chooses (risk/return combination) as long as the utility is the same.

To understand the indifference curve, let us plot all these numbers - expected return, standard deviation, and utility - on a graph.

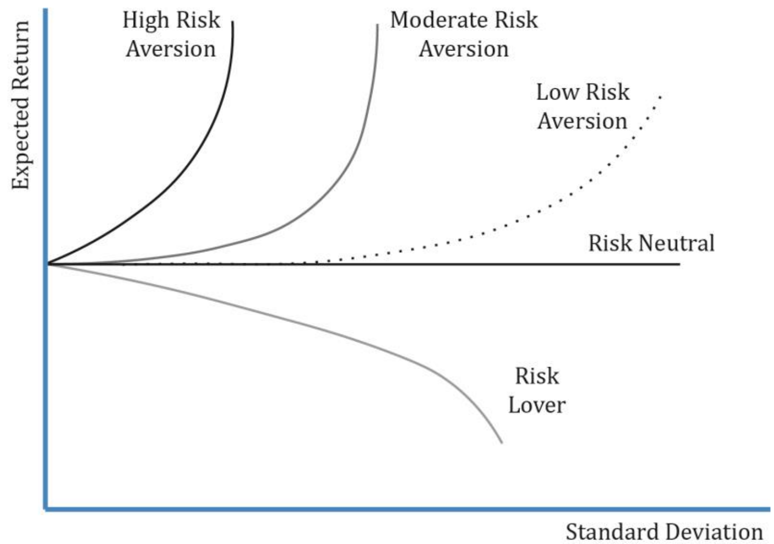


The key points to note are:

- The indifference curves run from south-west to north-east. This is intuitive because for taking on higher risk (going east), investors expect to be compensated with higher expected return (going north).
- The utility of points A and B are 0.05. This means that the investor will be indifferent about choosing an investment with E(r) of 5%, σ = 0 versus another with E(r) of 6%, σ = 10% as both have the same utility. Similarly, for assuming higher levels of risk, the investor is compensated with higher return as shown in the 0.05 utility curve. Along this curve, the investor is indifferent about choosing one point (investment) over the other.
- As the investor moves north-west, he is happier as his utility increases, in this case

from 0.05 to 0.06. Risk is compensated with higher returns and it signifies higher risk aversion.

We now consider the indifference curves for different types of investors.



The exhibit above shows the indifference curves for different types of investors with expected return on y-axis and standard deviation on x-axis. Note the following:

- Risk-neutral investor: For a risk-neutral investor, the utility is the same irrespective of risk as the investor is concerned only about the return. The expected return is constant.
- Risk-averse investor: For a risk-averse investor, the curve is upward sloping, as the investor expects additional return for taking additional risk. Finance theory assumes that most investors are risk averse, but the degree of aversion may vary. The curve is steeper for investors with high risk-aversion.
- Risk-seeking investor: The curve is downward sloping as the expected return decreases for higher levels of risk. It is not commonly seen.

6. Application of Utility Theory to Portfolio Selection

Now that we have seen utility theory and indifference curves for various investors, let us see how to apply it in portfolio selection. The simplest case is when a portfolio comprises two assets: a risk-free asset and a risky asset. For a high-risk averse investor, the choice is easy, to invest 100% in the risk-free asset but at the cost of lower returns. Similarly, for a risk-lover it would be to invest 100% in the risky asset. But is it the optimal allocation of assets, or can there be a trade-off?

Example

Consider a simple portfolio of a risk-free asset and a risky asset. Plot the expected return of the portfolio against the risk of the portfolio for different weights of the two assets.

Risk-free asset	Risky asset
R_f = 5%	R_i = 10%
σ = 0	σ_i = 20%

Solution:

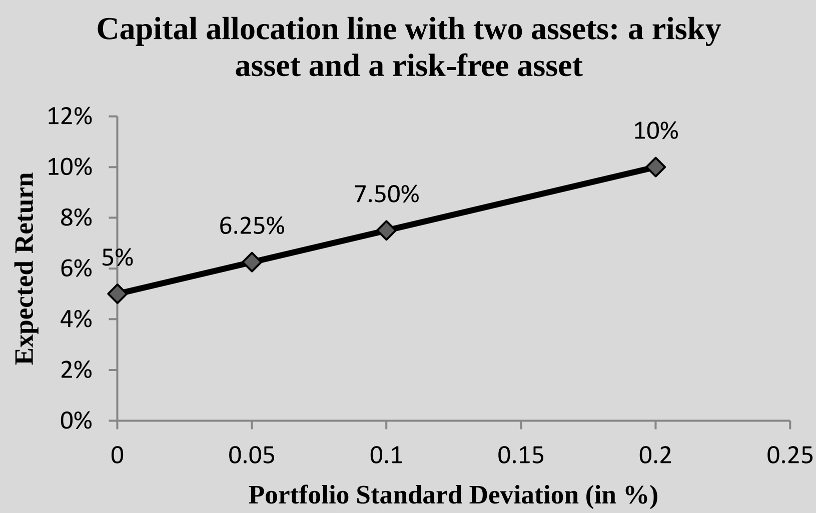
A portfolio's standard deviation is calculated as:

$$\sigma_p = \sqrt{w_1^2 * \sigma_1^2 + w_2^2 * \sigma_2^2 + 2 * w_1 * w_2 * \rho_{1,2} * \sigma_1 * \sigma_2}$$

Given that σ_1 = 0, the terms $w_1^2 * \sigma_1^2$ and $2 * w_1 * w_2 * \rho_{1,2} * \sigma_1 * \sigma_2$ become zero as the correlation between a risk-free asset with risky asset is zero.

So, the equation becomes $\sigma_p = w_2 * \sigma_2$. For different weights of the risky asset, let's calculate the values for risk and return and plot portfolio risk-return on a graph. Note that the expected return is a weighted average of the two assets.

Weight of risky asset = w_2	**σ_p (portfolio risk) = w_2 * σ_2**	**R_p (Portfolio return) = w_1* R_f + w_2 * R_i**
0	0	5%
0.25	5%	6.25%
0.5	10%	7.5%
1	20%	10%



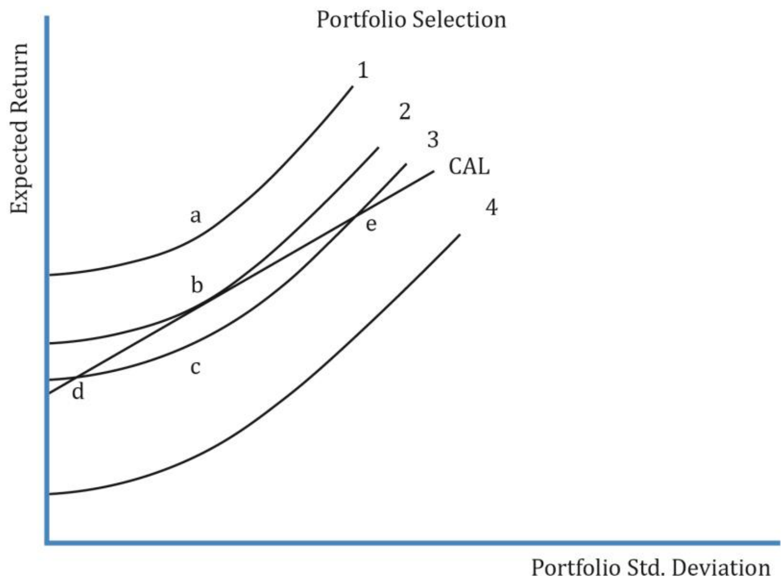
Graph interpretation:

- What you get by plotting risk and return values for different weights of the risky and risk-free asset is a straight line (linear).
- As you move north-east, the weight of the risky asset increases. The lowest point on the left corner, intersecting the y-axis with expected return of 5%, represents 100% in the risk-free asset. Similarly, the topmost point on the right corner represents 100% in the risky asset with expected return of 10%.

- The straight line you see in the chart above is called the capital allocation line which represents the portfolios available to the investor as each point on the line is an investable portfolio.
- So, how does an investor decide which portfolio to invest in? The answer is by combining an investor's indifference curves and CAL to determine the optimal portfolio. We look at it in detail in the next section.

What is the Optimal Portfolio?

We get the optimal portfolio by combining the indifference curves and the capital allocation line. The utility theory gives us the indifference curves for an individual, while the capital allocation lines give us a set of feasible investments. The optimal portfolio for an investor will lie somewhere on the capital allocation line, i.e., some combination (weight) of the risky asset and risk-free asset. Using an investor's risk-aversion measure A, we can use the utility theory to plot the indifference curves for an individual. Assume the indifference curves for this investor look as shown in the exhibit below:



Some key points about the graph:

- Curves 1, 2, 3, and 4 are various indifference curves for an individual. The indifference curves cannot intersect each other.
- Curve 1 lies above the CAL. Curve 2 is tangential to the CAL at point *b*. Curve 3 intersects CAL at two points: *d* and *e*. Curve 4 lies below the CAL.
- Curve 1 has the highest utility while Curve 4 has the lowest utility. As we discussed before, the investor is happier when we move north-west. Of the four curves, he will be happiest with curve 1. But, as curve 1 lies outside the CAL, it is not possible to construct a portfolio at any point on this curve with the available assets.

- Points on curve 3 may be attainable with the two assets. To achieve the utility level of curve 3, the investor may choose to invest at points d and e where it intersects with CAL.
- However, curve 2 is preferred over curve 3, as it provides a higher utility. To achieve this utility level, the investor may choose to invest at point b, where the indifference curve is tangential to the capital allocation line
- This point, b, represents the optimal portfolio.

If another investor has a higher level of risk aversion, where will his optimal portfolio lie? With a higher risk aversion level, the indifference curves will be steeper. The tangential point will be closer to the risk-free asset, so it will be to the south-west of point b.

7. Portfolio Risk & Portfolio of Two Risky Assets

Portfolio risk depends on:

- Risk of individual assets.
- Weight of each asset.
- Covariance or correlation between the assets.

Portfolio of Two Risky Assets

Portfolio Return

A portfolio is usually composed of more than one asset in different proportions. Portfolio return is the weighted average of the returns of the individual investments.

Assume you have three different stocks A, B, and C in your portfolio with weights of 50%, 25%, and 25% respectively. The returns on A, B, and C are 20%, 10%, and 10% respectively. The portfolio return is the weighted average return of three stocks: $0.5 \text{ x } 20 + 0.25 \text{ x } 10 + 0.25 \text{ x } 10 = 15\%$.

Risk of a two-asset portfolio is given by:

$$\sigma_p = \sqrt{w_1^2 \text{ x } \sigma_1^2 + w_2^2 \text{ x } \sigma_2^2 + 2 \text{ x } w_1 \text{ x } w_2 \text{ x } \rho_{1,2} \text{ x } \sigma_1 \text{ x } \sigma_2}$$

$$\sigma_p = \sqrt{w_1^2 \text{ x } \sigma_1^2 + w_2^2 \text{ x } \sigma_2^2 + 2 \text{ x } w_1 \text{ x } w_2 \text{ x } Cov_{1,2}}$$

Covariance $= Cov_{1,2} = \rho_{1,2} \text{ x } \sigma_1 \text{ x } \sigma_2$

where: $\rho_{1,2}$ = correlation coefficient that gives the correlation between returns R_1 and R_2.

Impact of correlation on portfolio risk

As correlation decreases, the diversification benefit increases, i.e., the risk of the portfolio decreases.

Example

Let us revisit our portfolio of X and Y. Earlier, we considered the impact of changing weights of the two assets. Now, we will also consider the impact of different correlations. The relevant data is reproduced here. X has an expected return of 12% and a standard deviation of 16%. Y has an expected return of 20% and a standard deviation of 30%. What is the expected return and risk of the portfolio for different weights assuming the following correlations: -1, -0.5, 0, 0.5, and 1?

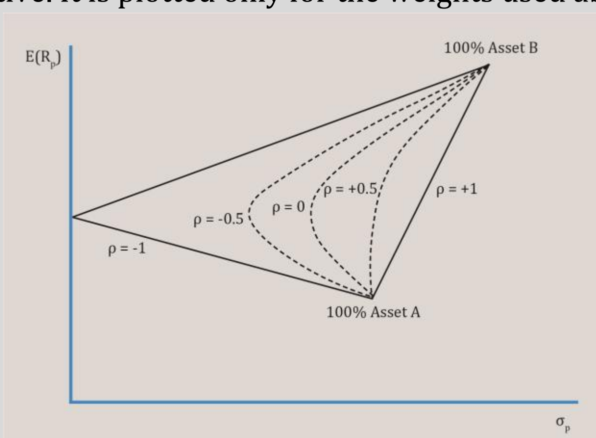
Solution:

$R_p = w_x R_x + w_y R_y$ $\sigma_p^2 = w_x^2\sigma_x^2 + w_y^2\sigma_y^2 + 2w_xw_y\rho_{xy}\sigma_x\sigma_y$

R_x = 12%, σ_x = 16%; R_y = 20%; σ_y = 30%

Weight of X (%)	**Portfolio return**	Portfolio risk when correlation between X and Y is:				
		-1	**-0.5**	**0**	**0.5**	**1**
0	20	30	30	30	30	30
20	18.4	20.8	22.5	24.2	25.7	27.2
50	16	7	13	17	20.2	23
60	15.2	2.4	10.9	15.4	18.7	21.6
70	14.4	2.2	10.28	14.3	17.5	20.2
100	12	16	16	16	16	16

This chart is not exhaustive. It is plotted only for the weights used above.



- Consider the line represented with a correlation of 1. As the weight of X is reduced and weight of Y is increased, the risk and return both increase. There is no diversification benefit.
- Consider a correlation of 0. As the weight of X is reduced and weight of Y is increased, the risk initially decreases and the return increases. Hence there is clearly a diversification benefit.
- The diversification benefit increases as correlation decreases.
- With a correlation of -1, there is a certain weight of X and Y when the risk is zero.

8. Portfolio of Many Risky Assets

In the previous example, we saw how a two-asset portfolio's risk reduces as the correlation between the assets decreases. To be more generic, for a portfolio with more than two assets, the variance is given by:

$$\sigma_P^2 = \frac{\sigma^2}{N} + \frac{N-1}{N} * \text{Cov}$$

where:
σ_P^2 = portfolio variance
σ^2 = average variance
Cov = average covariance
N = number of assets in the portfolio

Instructor's tip:
The probability of being tested on this formula is low.

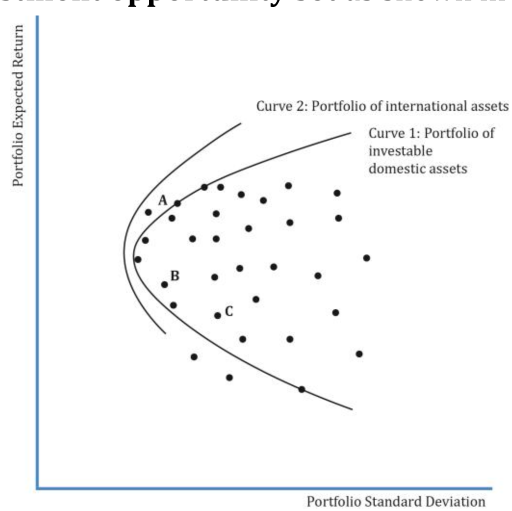
9. The Power of Diversification

When creating a diversified portfolio, the choice of assets and the correlation among assets is important. Adding perfectly positively correlated assets to a portfolio does little to reduce the portfolio's risk. Similarly, assets that have lower correlations will reduce risk. When two assets with a correlation of -1.0 are combined to form a portfolio, risk can be reduced to zero.

10. Efficient Frontier: Investment Opportunity Set & Minimum Variance Portfolios

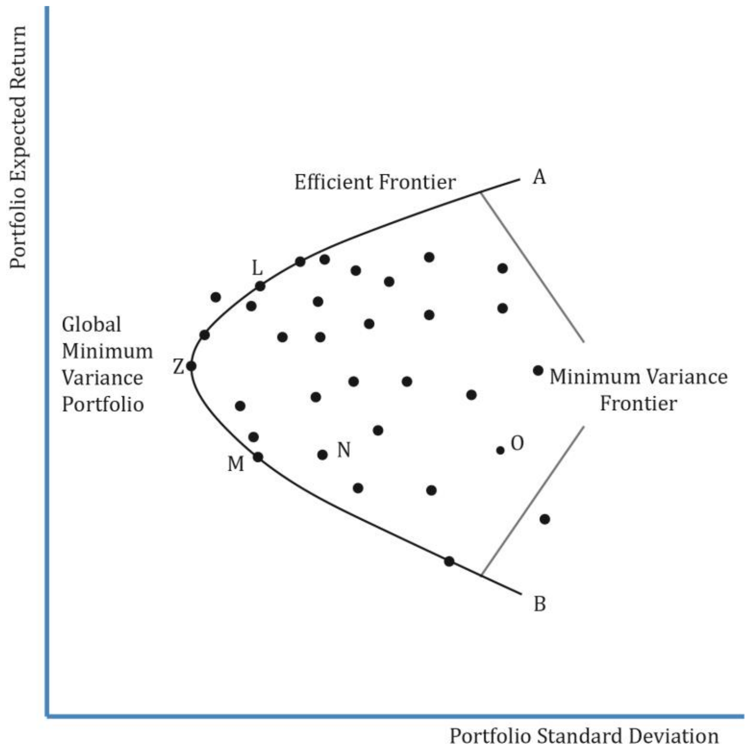
Investment Opportunity Set

Consider the universe of risky, investable assets available to an investor. These can be combined to form many portfolios. When plotted together they form a curve. This set of portfolios is called the **investment opportunity set** as shown in the exhibit below.



Minimum Variance Portfolios

At any given level of return in the investment opportunity set, there will be a portfolio with minimum variance, i.e., minimum risk for a given level of return. The line combining such portfolios with minimum variance is called the **minimum-variance frontier**.



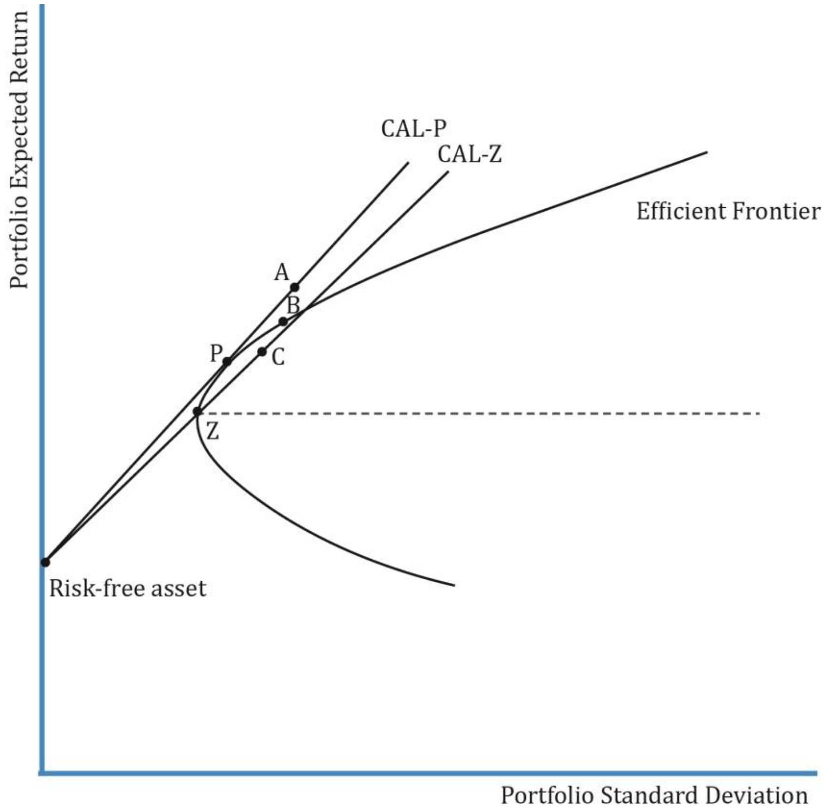
In the exhibit above, points M, N and O have the same expected return. But, investors will prefer point M over N or O because it has lower risk. The curve stretching from A to B through Z is called the **minimum-variance frontier.**

Global Minimum-Variance Portfolio: the portfolio with the lowest risk or minimum variance among portfolios of all risky assets is called the global minimum-variance portfolio. It is represented by point Z in the exhibit above. It is not possible to hold a portfolio of risky assets that has less risk than that of the global minimum-variance portfolio.

The **efficient frontier** is the part of the minimum variance frontier that is above the global minimum-variance portfolio. It represents the set of portfolios that will give the highest return at each risk level. *Remember that the efficient frontier consists of only risky assets. There is no risk-free asset.* Consider points L and M on the minimum-variance frontier. Both the portfolios have the same level of risk but portfolio L has a higher return than M. Investors will prefer the portfolio with higher return.

11. Efficient Frontier: A Risk-Free Asset and Many Risky Assets

One of the constraints of the efficient frontier is that it comprises only risky assets. We overcome this drawback with the capital allocation line: a combination of risky assets and a risk-free asset. The addition of a risk-free asset makes the investment opportunity set much richer than the investment opportunity set consisting only of risky assets. A portfolio of risky assets and a risk-free asset results in a straight line represented by the **capital allocation line**.

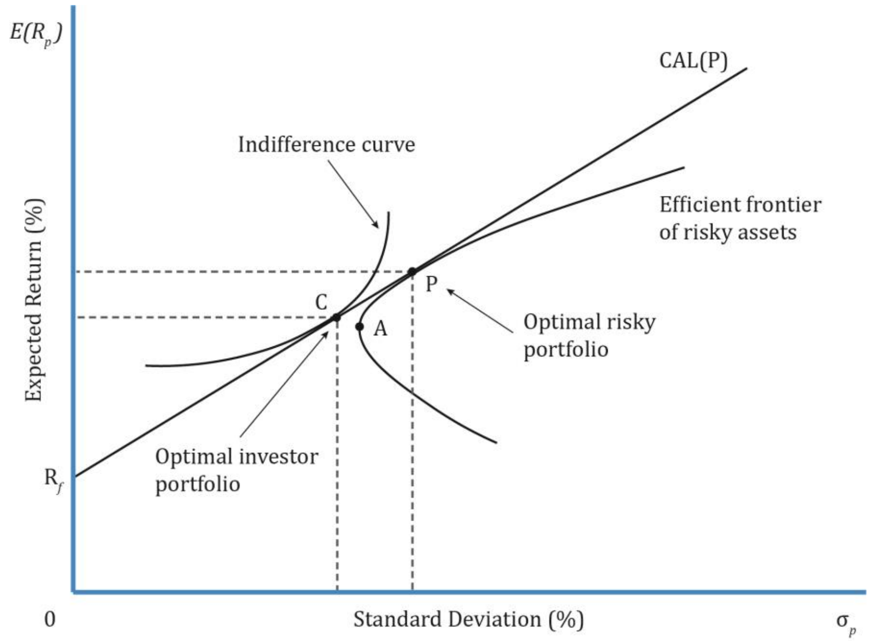


Interpretation of the diagram:

- CAL-P and CAL-Z are capital allocation lines formed by combining risky portfolios P and Z with the risk-free asset respectively.
- Portfolio P is the **optimal risky portfolio.** It is the point at which the capital allocation line is tangential to the efficient frontier of risky assets. CAL-P is the optimal capital allocation line.
- The optimal risky portfolio is based on the market and not the investor's risk preferences. Along CAL-P, the weight of risk-free assets and portfolio P varies. For instance, it is 100% in the risk-free asset at y-axis, 100% in risky portfolio P at the point P, and a mix of both in between.

12. Efficient Frontier: Optimal Investor Portfolio

To identify the optimal investor portfolio, we must consider the investor's risk preferences. The utility of each investor is best represented by his indifference curves. The optimal investor portfolio is the point at which the investor's indifference curve is tangential to the optimal allocation line.



Portfolio C in the exhibit above is the optimal investor portfolio.

Summary

LO: Describe characteristics of the major asset classes that investors consider in forming portfolios.

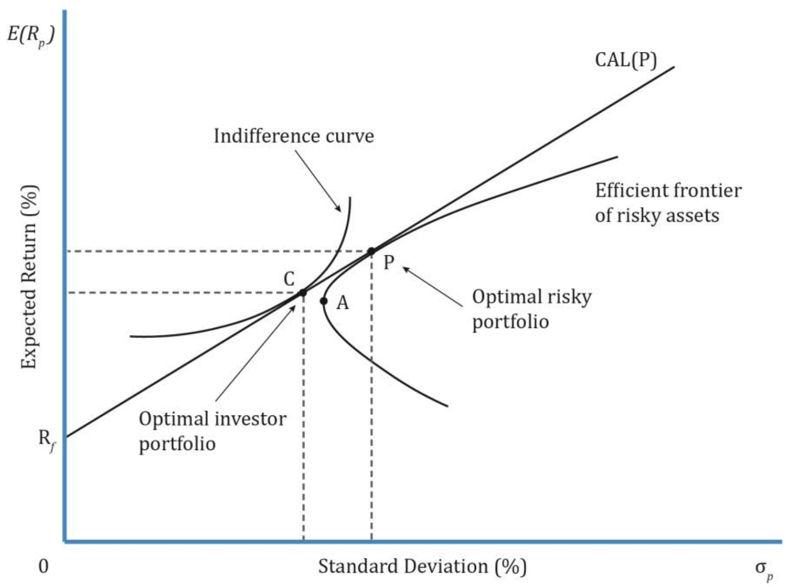
Asset Class	Annual Average Return	Standard Deviation (Risk)
Small-cap stocks	High	High
Large-cap stocks	↓	↓
Long-term corporate bonds		
Long-term treasury bonds		
Treasury bills	Low	Low

LO: Explain risk aversion and its implications for portfolio selection.

A risk-averse investor prefers less risk to more risk. In contrast, a risk seeking investor prefers more risk to less risk. If expected returns of two assets are same, a risk-averse investor will prefer asset with lower risk.

LO: Explain the selection of an optimal portfolio, given an investor's utility (or risk aversion) and the capital allocation line.

To identify the optimal investor portfolio, we must consider the investor's risk preferences. The utility of each investor is best represented by his indifference curves. The optimal investor portfolio is the point at which the investor's indifference curve is tangential to the optimal allocation line. Portfolio C in the exhibit below is the optimal investor portfolio.



CAL (P) is based on the market, not the investor's risk preferences; it represents the most efficient portfolio for each level of risk.

LO: Calculate and interpret the mean, variance, and covariance (or correlation) of asset returns based on historical data.

Variance is a measure of volatility or dispersion of returns for a single variable.

Covariance is a measure of how two variables move together. It is difficult to interpret.

Correlation is a standardized measure of the linear relationship between two variables with values ranging between -1 and +1. It can be calculated using the formula:

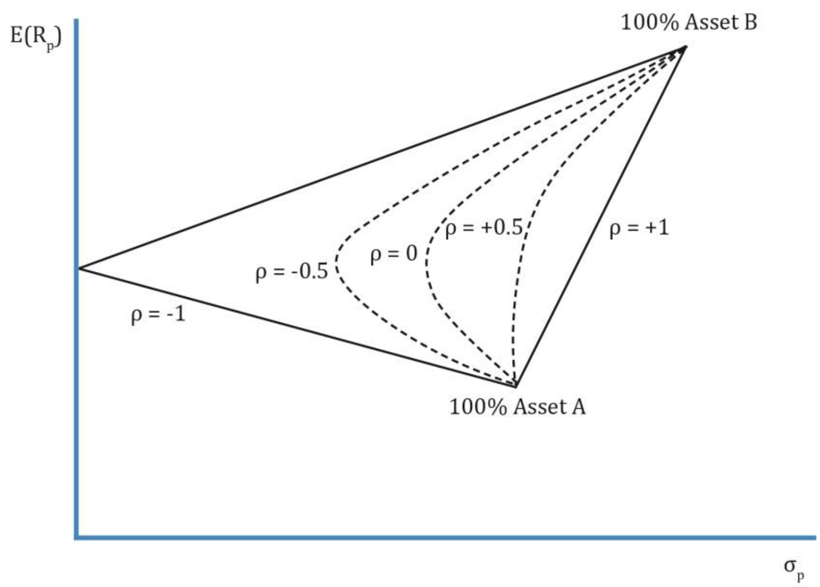
$$\rho(Ri, Rj) = Cov\ (Ri, Rj)/\sigma(Ri)\sigma(Rj)$$

LO: Calculate and interpret portfolio standard deviation.

The standard deviation of a portfolio of two risky assets can be calculated using the following formula:

$$\sigma_p = \sqrt{w_1^2 * \sigma_1^2 + w_2^2 * \sigma_2^2 + 2 * w_1 * w_2 * Cov_{1,2}}$$

LO: Describe the effect on a portfolio's risk of investing in assets that are less than perfectly correlated.



The above graph of risk and return shows the benefits of diversification. When the correlation of assets in a portfolio decreases, the risk of the portfolio decreases.

LO: Describe and interpret the minimum-variance and efficient frontiers of risky assets and the global minimum-variance portfolio.

Consider the universe of risky, investable assets. These can be combined to form many portfolios; when plotted together they form a curve. This set of portfolios is called the investment opportunity set.

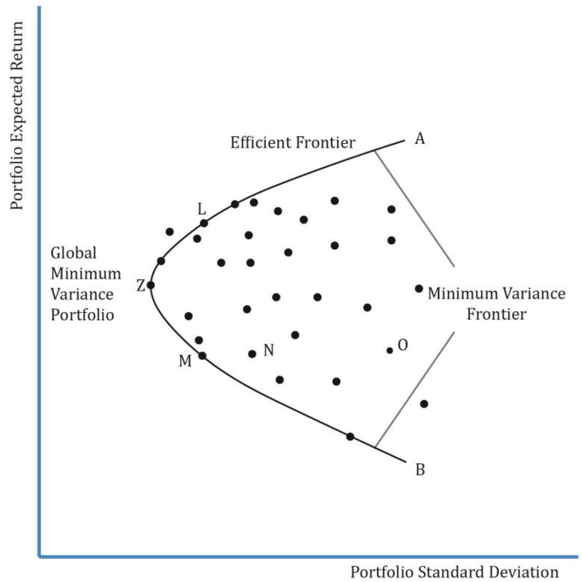
At any given level of return in the investment opportunity set, there will be a portfolio with minimum variance. The line combining such portfolios with minimum variance is called the minimum-variance frontier.

The portfolio with the lowest risk or minimum variance among portfolios of all risky assets is called the global minimum-variance portfolio.

The efficient frontier is the part of the minimum-variance frontier that is above the global minimum-variance portfolio. It represents the set of portfolios that will give the highest return at each risk level.

Remember that the efficient frontier consists of only risky assets. There is no risk-free asset.

The graph below shows these points:



LM03 Portfolio Risk and Return: Part II

Version 1.0

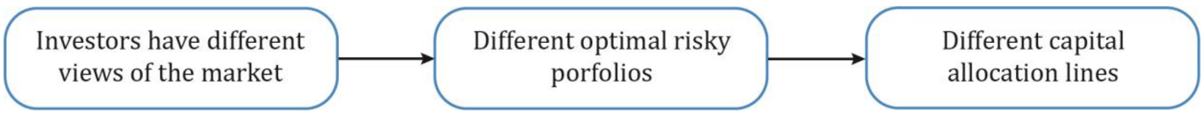
1. Introduction

In this reading, we will discuss:

- The capital market line (CML).
- The two components of total risk: systematic and nonsystematic risk.
- The capital asset pricing model (CAPM) and the security market line (SML). The primary objective of this reading is to identify the optimal risky portfolio using CAPM.

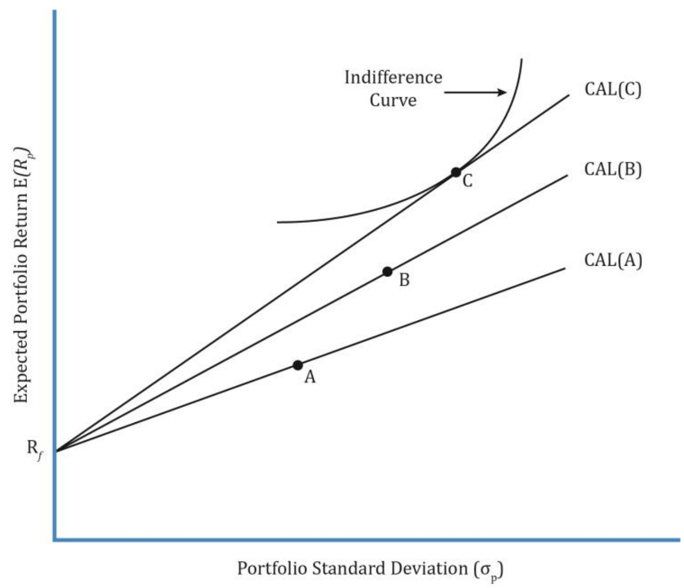
2. Capital Market Theory: Risk-Free and Risky Assets

Portfolio of Risk-Free and Risky Assets



This is a repetition of what we have seen in the previous reading. When a risk-free asset is combined with a risky asset, it results in higher risk-adjusted returns because the risk-free asset has zero correlation with the risky asset. This leads to the capital allocation line.

Investors have different views of the market (and different levels of risk aversion) which means the individual risky assets (e.g. securities) they choose to form their portfolio are different. Combining the capital allocation line with an investor's indifference curve leads to different optimal risky portfolios, as illustrated in the exhibit below.



We now explore whether a unique optimal risky portfolio exists for all investors. The answer is yes, if there is homogeneity of expectations.

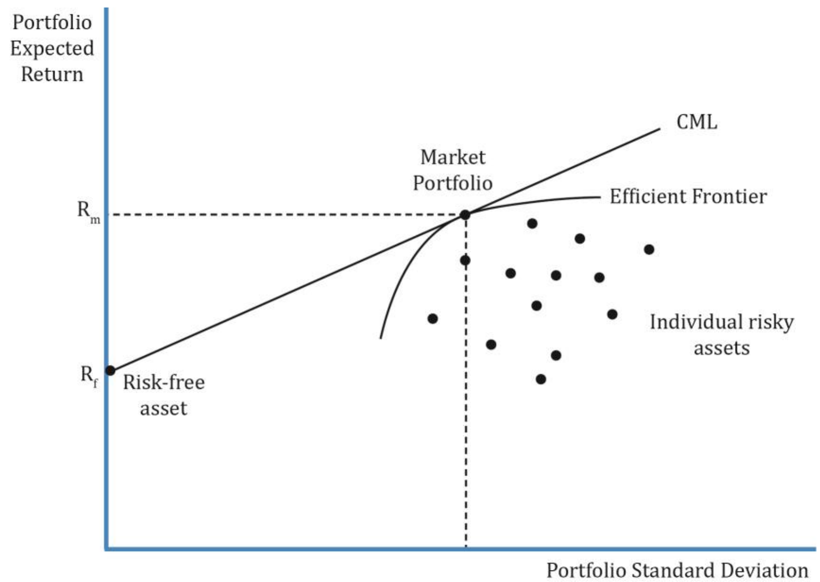
What is homogeneity of expectations?

- It means that all investors have the same economic expectations for an asset.

- For example, assume there are two securities. With homogeneity of expectations, all investors have the same views on risk, return, price, cash flows, and the correlation between the two assets. This means the calculation to arrive at optimal risky portfolio for all investors would be the same.

3. Capital Market Theory: The Capital Market Line

Since all investors have the same expectations, they will construct only one efficient frontier. If there is one efficient frontier, there will be only one capital allocation line. The point where this capital allocation line is tangential to the efficient frontier is called the market portfolio. This is the optimal risky portfolio when all investors have the same expectations. The CML is a special case of the CAL where the efficient portfolio is the market portfolio.



Now, let us derive the equation for CML. We will use a basic equation of the form: Y = C + mX

where:

Y= R_p (portfolio return)
C = R_f (risk-free rate)
m = slope = $(R_m - R_f) / \sigma_m$
X = portfolio standard deviation = σ_p

The equation for CML can be written as

$$R_p = R_f + \left(\frac{R_m - R_f}{\sigma_m}\right) * \sigma_p$$

Any point along the CML is a combination of the risk-free asset and the market portfolio. At the point where the CML intersects y-axis, 100% is invested in the risk-free asset and its weight decreases as we go up along the CML.

What is the market?

Theoretically, the market includes all risky assets or anything that has value. Examples include stocks, bonds, real estate, etc.

The market portfolio should contain as many assets as possible, but it is not practical to include all assets in a single risky portfolio because not all assets are tradable and not all tradable assets are investable. Examples of non-tradable assets include the Eiffel Tower and human capital. Examples of tradable assets that are not investable include Class A shares on the Shanghai Stock Exchange that are not available for foreign investors.

Practically, we use major stock market indexes as a proxy for the market. This reading uses the S&P 500 index as the market's proxy as it represents approximately 80% of the U.S stock market capitalization and U.S. markets account for nearly 32 per cent of the world markets.

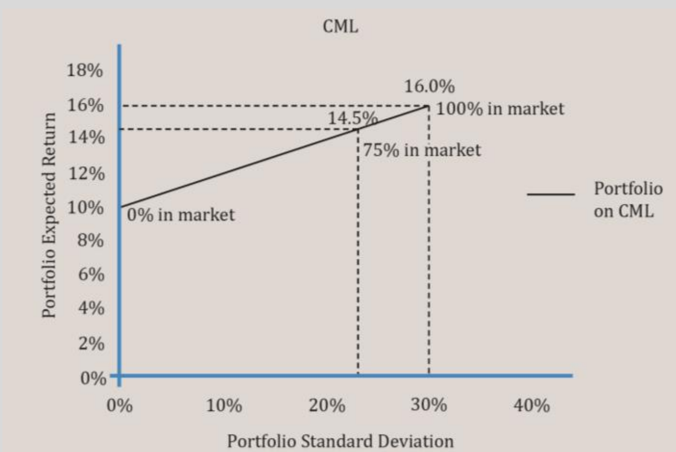
Example

Majid Khan wants to build a portfolio using T-bills and an index fund that tracks the KSE 100 Index. The T-bills have a return of 10%. The index fund has an expected return of 16% and a standard deviation of 30%. Draw the CML. Show the point where the investment in the market is 75%. What is the risk and return at this point?

Solution:

Risk and return when the investment in the market is 75%:
$\sigma_p = w_m * \sigma_m$ = 0.75 * 30 = 22.5% (*Note: Risk of T-bills is zero*)
$R_p = w_{r_f} \text{ x } R_f + w_m \text{ x } R_m$ = 0.25 x 10 + 0.75 x 16 = 14.5%

Let us now plot these values on the CML.

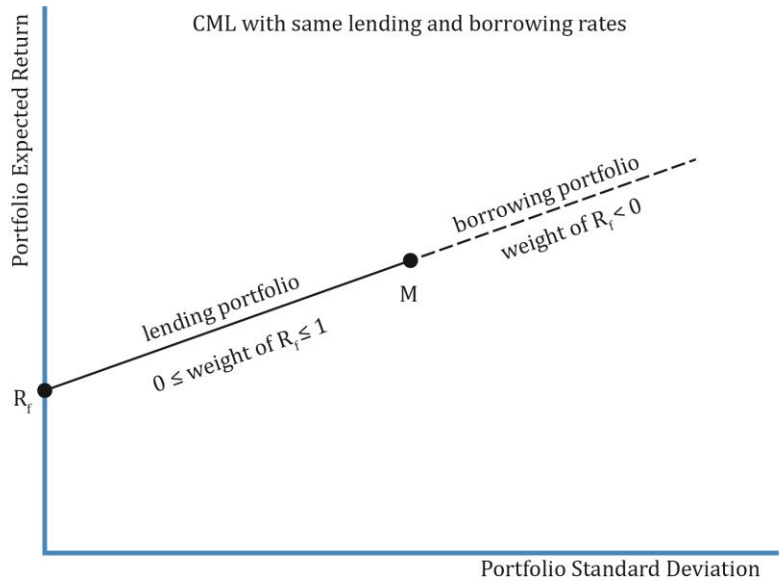


When the return on the portfolio is 10%, Majid is holding 100% of the T-bills and 0% of the market portfolio. As he increases his weight in the market portfolio to 75%, we see that the return has changed to 14.5% but his risk has also gone down to 22.5%. When he has invested 100% in the market portfolio, the risk and return are the same as the market portfolio.

4. Capital Market Theory: CML - Leveraged Portfolios

Our focus so far has been on the portfolios between the points R_f, the risk-free asset, and M, the market portfolio on the CML, where an investor is holding some combination of the two assets. The portfolios on this stretch between R_f and M are called lending portfolios because the investment in the assets comes from the investor's own wealth. He is lending (investing) his wealth at the risk-free rate.

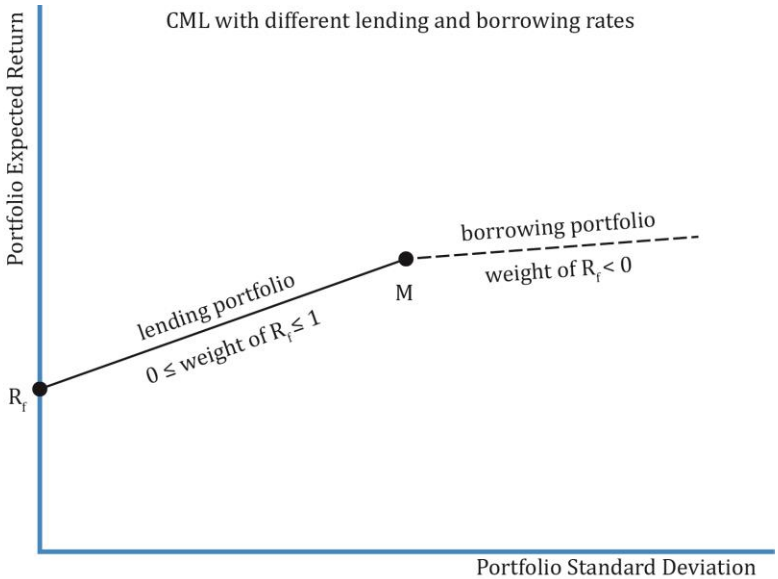
The dotted line in the exhibit below stretches beyond M on the CML represents the borrowing portfolios. This means that the investor is able to *borrow* money at the risk-free rate to invest more in the market portfolio.



- A portfolio beyond point M is a combination of the investor's own wealth and borrowed money at the risk-free rate.
- As we go up north-east beyond M on the CML, the investment in the market portfolio increases and so does the amount of borrowed money.
- Beyond point M, there is a negative investment in the risk-free asset, and the risky portfolio is called the leveraged portfolio. More risk leads to higher expected return.
- How much money the investor borrows to invest in a portfolio beyond M depends on his risk-return preferences, in other words, his utility function.
- The slope of the line between R_f and M is given as $\frac{R_m - R_f}{\sigma_m}$.

Different lending and borrowing rates

Often, investors are not able to borrow money at the risk-free rate. If the borrowing rate is higher than the risk-free rate, the CML is no longer a straight line. The slope of the line to the right of M is given by $\frac{R_m - R_b}{\sigma_m}$. Hence, there is a 'kink' in the CML. This is illustrated in the exhibit below. Using the data from the previous example, the slope of CML when the borrowing rate is the same as lending rate, we get slope as $\frac{16-10}{30} = 0.2$. Now, assume the borrowing rate is higher at 12%. The slope in this case will be $\frac{16-12}{30} = 0.133$, which is less than 0.2.



When the borrowing rate is higher than the lending rate, the return per unit of risk is less, relative to when the borrowing rate is equal to the lending rate.

Example

After a successful initial foray into the stock market, Majid Khan gets a little greedy and decides to build a leveraged portfolio. He invests 100,000 of his own money and 50,000 of borrowed money in the index fund. He expects to pay 10% on the borrowed money. The index fund has an expected return of 16% and a standard deviation of 30%. Calculate his expected risk and return.

Solution:

w_1 = Weight of the risk-free asset = Weight of borrowed money = -50,000/100,000 = -0.5

w_2 = Weight in risky asset = 150,000/100,000 = 1.5

Total weight = $w_1 + w_2$ = 1.5 - 0.5 = 1

Expected return with -50% invested in T-bills = $w_1R_b + w_2R_m$ = -0.5 (10) + 1.5 (16) = 19%.

Standard deviation with -50% invested in T-bills is $w_2 \times \sigma_m$ = 1.5 x 30 = 45%.

This example shows us that leverage amplifies both expected return and risk.

Example

Calculate the expected return and risk if Majid actually needs to pay 12% on the borrowed money. All other numbers are the same.

Solution:

The standard deviation or risk of the portfolio remains the same. But the expected return must be lower as the borrowing rate has increased. Let us calculate the slope of the borrowing part of the CML as $\frac{R_m - R_b}{\sigma_m} = \frac{16 - 12}{30} = 4/30$.

Expected return = 12% + (16 - 12)/30 * 45 = 18%.

Or, expected return with -50% invested in T-bills = $w_1R_b + w_2R_m$ = -0.5 (12) + 1.5 (16) = 18%.

Standard deviation with -50% invested in T-bills $\sigma_p = w_2 * \sigma_M$ =1.5 * 30 = 45%.

5. Systematic and Nonsystematic Risk

We have seen repeatedly that high returns come with high risk. But does all high risk lead to high returns? No. Total risk can be decomposed into systematic and nonsystematic risk.

Total variance = Systematic variance + Nonsystematic variance

Systematic Risk and Nonsystematic Risk

Systematic risk is non-diversifiable or market risk that affects the entire economy and cannot be diversified away. Investors get a return for systematic risk.

Examples of systematic risk are interest rates, inflation, natural disasters, unrest/coup attempts such as events in Middle East/Africa since 2011 and political uncertainty.

It is tough to avoid systematic risk. However, the effects can be minimized by selecting securities with low correlation to the rest of the portfolio.

Nonsystematic risk is a local risk that affects only a particular asset or industry. There is no compensation for nonsystematic risk as it can be diversified away.

Examples of nonsystematic risk are oil discoveries, non-approval for a drug, new regulations for telecom industry.

Nonsystematic risk may be avoided by diversifying a portfolio to include assets across countries, industries, and asset classes. Nonsystematic risk does not earn a return.

Pricing of risk**:** Pricing an asset is equivalent to estimating its expected rate of return; price and return are often used interchangeably. For example, if we know the cash flows for a bond, its price can be computed using the discount rate. Pricing of risk is this rate of return which reflects the systematic risk.

Instructor's Note
Systematic risk or market risk is the only risk for which investors get compensated. It cannot be eliminated with diversification.

A risk-free asset has zero systematic risk and zero nonsystematic risk.

Example

Describe the systematic and nonsystematic risk components of the following assets:

- A six-month Treasury bill.
- An index fund based on the S&P 500, with a total risk of 18%.

Solution:

- A six-month Treasury bill: For a T-bill, we know how much an investor would get paid at the end of six months. The systematic risk and nonsystematic risk are both zero; the total variance is therefore zero.
- An index fund based on the S&P 500, with a total risk of 18%: S&P 500 represents the market. There is no nonsystematic risk in a market portfolio. So, the total risk of 18% is systematic risk.

Example

Consider two assets X and Y. X has a total risk of 25% of which 15% is systematic risk. Asset Y is the S&P index fund mentioned above with a total risk of 18%, all of which is systematic risk. Which asset will have a higher expected rate of return?

Solution:

Based on total risk, asset X seems an obvious choice because higher risk results in higher return. However, it is more appropriate to compare the systematic risk of the two assets as only systematic risk earns a return. Asset Y will earn a higher expected return as its systematic risk of 18% is higher than asset X's 15%. An investor will not be compensated for assuming nonsystematic risk in asset X as it can be eliminated.

6. Return-Generating Models

Return-generating models provide an estimate of the expected return of a security given certain input parameters called factors. If the model uses many factors, it is called a multi-factor model. If it uses one factor, it is a single-factor model.

<u>Multi-factor Models</u>

The three commonly used multi-factor models are: macroeconomic, fundamental, and statistical models. These models use factors that are correlated with security returns.

- Macroeconomic models use economic factors such as economic growth, interest rate, unemployment rate, productivity, etc.
- Fundamental models use input parameters such as earnings, earnings growth, cash flow, market capitalization, industry-specific inputs, financial ratios like P/E, P/B, value/growth, etc.
- In the statistical models, there is no observable economic or fundamental connection between the input and security returns. Unlike a macroeconomic or fundamental model, there are also no pre-determined set of factors. Instead, historical or cross-

sectional security returns are analyzed to identify factors that explain variance or covariance in returns.

Single-factor Models

In a single-factor model, only one factor is considered. A classic single factor model is the market model which is given by this equation:

$$R_i = \alpha_i + \beta R_m + e_i$$

where R_i is the dependent variable and R_m is the independent variable.

According to the market model, a stock's return can be decomposed into:

- α_i – Excess returns of the stock (difference between expected and realized returns).
- β – Systematic risk of a security or the stock's sensitivity to the market. For instance, how the stock's return varies when the market return moves up or down by 1%.
- R_m – Return on the market.
- e_i – Error term that affects stock returns due to firm-specific factors. There is an error term because not all of a stock's returns can be explained by market returns.

7. Calculation and Interpretation of Beta

Beta is a measure of systematic (or market) risk. Beta is a number and has no unit of measure. It tells us how sensitive an asset's return is to the market as a whole. Beta determines the amount by which a stock's return is expected to move for every 1 per cent increase or decrease in the market return. Beta is the ratio of the covariance of returns of stock i with returns on the market to the variance of market return.

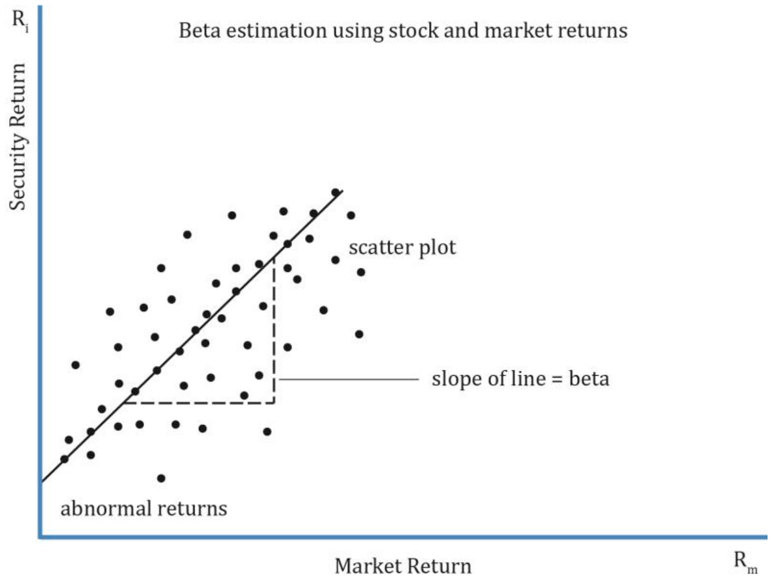
$$\beta = \frac{Cov(i,M)}{\sigma_M^2} = \frac{\rho_{iM} * \sigma_i}{\sigma_M}$$

Interpretation of beta values:

- **β > 0:** Return of the asset follows the market trend.
- **β < 0:** Return of the asset moves in an opposite direction to the market trend (negatively correlated with the market).
- **β = 0:** An asset's return has no correlation with the market. For example, a risk-free asset has a beta of zero.
- **β = 1:** Beta of market is equal to 1. If a stock has a beta of 1, it means it has the same volatility as that of the market.

How do we estimate beta?

Beta is estimated using a statistical method called linear regression analysis where stock returns are regressed against market returns. Historical security returns and historical market returns are used as inputs in this method. Let us say we consider a stock's returns for the past 100 months. The pairs (security, market return) are plotted to get a scatter plot. Each point on the graph represents a month. A regression line is drawn though the scatter plot that best represents the data points. The point where the line intercepts the y-axis is α (or excess returns). The slope of the line is equal to its beta.



A portfolio beta is calculated as the weighted average beta of the individual securities in the portfolio.

Example

Assuming the standard deviation of market returns is 20%, calculate the beta for the following assets:

- 6-month T-bill.
- A commodity with σ = 15%, and zero correlation with the market.
- A stock with σ = 25% and correlation with the market = 0.6.

Solution:

- 6-month T-bill: beta for a T-bill is zero as there is no correlation between a risk-free asset and the market.
- A commodity with σ = 15%, and zero correlation with the market: using equation 7, we can conclude that beta is zero if correlation with the market is zero.
- A stock with σ = 25% and correlation with the market = 0.6: $\beta = \frac{0.6 * 0.25}{0.2} = 0.75$

8. Capital Asset Pricing Model: Assumptions and the Security Market Line

The capital asset pricing model is a model used **to calculate the expected rate of return of a risky asset**, and gives the relationship between a security's return and risk. CAPM states that the expected return of assets reflects only their systematic risk, which is measured by beta. Systematic or market risk cannot be diversified. It means that two assets with same beta will have the same expected return irrespective of their individual characteristics.

$$r_e = R_f + \beta\,[E\,(R_{mkt}) - R_f]$$

where:
r_e = expected return
R_f = risk − free rate
β = systematic risk of security
$E(R_{mkt})$ = expected return of market
$E(R_{mkt}) - R_f$ = market risk premium; it is the compensation for assuming market risk.

Assumptions of the CAPM

We will now take a look at the assumptions for arriving at CAPM:

1. Investors are risk-averse, utility-maximizing, rational individuals.
   Risk aversion means investors expect a higher return to accept a higher level of risk. Different investors have different levels of risk aversion or risk tolerance. Utility maximizing means investors prefer higher returns to lower returns. Individuals are rational means that all investors have the ability to gather and process information to make logical decisions. In reality, however, this is not true for most investors as personal biases and experiences shroud their judgment.
2. Markets are frictionless. There are no transaction costs and no taxes.
   Investors can borrow and lend as much as they want at the risk-free rate. One important assumption here is that there are no costs or restrictions on short-selling, which is not true in reality. This limits the CAPM.
3. All investors plan for the same, single holding period.
   CAPM is based on a single period instead of multiple periods because it is easy to calculate.
4. Investors have homogenous expectations.
   All investors analyze securities using the same probability distribution to arrive at identical valuations. This means all investors use the same estimates for expected returns, variance, and correlation between assets. Since the expected returns and standard deviation for all investors is the same, they will arrive at the same optimal risky portfolio, or the market portfolio in the CML.
5. Investments are infinitely divisible.
   Investors can hold a fraction of any asset. It means that they may invest as much as or little in any asset.
6. Investors are price takers.
   CAPM assumes that there are many small investors who cannot influence the security prices. They are price takers.

Example

FFC's standard deviation of returns is 25% and its correlation with the market is 0.6. The standard deviation of returns for the market is 20%. The expected market return is 10% and the risk free rate is 3%. What is FFC's expected return?

Solution:

$\beta = \frac{\rho_{iM} * \sigma_i}{\sigma_M} = \frac{0.6 * 0.25}{0.2} = 0.75$

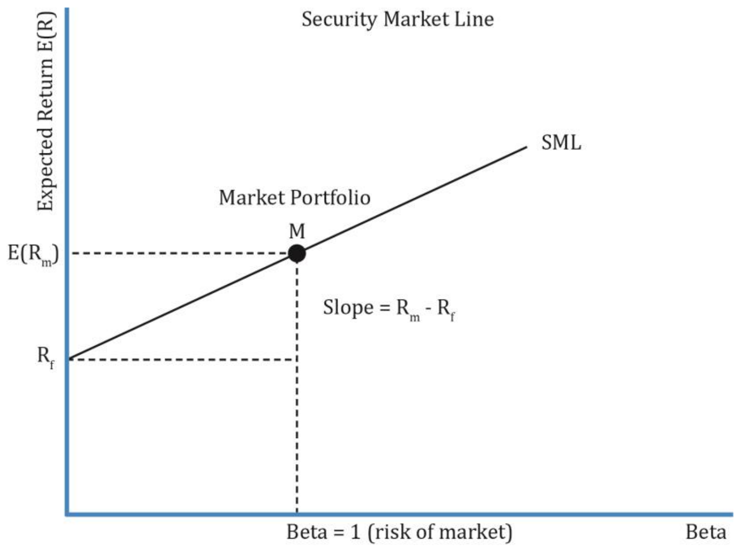
$r_e = R_f + \beta[E(R_{mkt}) - R_f]$

$r_e = 0.03 + 0.75[0.1 - 0.03] = 0.0825$

$r_e = 8.25\%$

The Security Market Line

The security market line (SML) is a graphical representation of the capital asset pricing model and applies to all securities, whether they are efficient or not. The graph has beta on the x-axis and expected return on the y-axis.



SML intersects y-axis at the risk-free rate.

Slope of the line = $\frac{R_m - R_f}{\beta} = R_m - R_f$ as β = 1 for the market. So, slope of the line is the market risk premium.

What is the difference between CML and SML?

Differences between CML and SML	
CML	**SML**
Does not apply to all securities; applies to only efficient portfolios.	Applies to any security. It may include inefficient portfolios as well.

Only systematic risk is priced. So, CAL/CML can only be applied to those portfolios whose total risk is equal to systematic risk.	
X-axis has the standard deviation of the portfolio.	X-axis has beta of the asset or portfolio.
$R_p = R_f + (\frac{R_m - R_f}{\sigma_m}) * \sigma_p$ where σ_p is the total risk of the portfolio.	$r_e = R_f + \beta[E(R_{mkt}) - R_f]$ where β is the systematic risk. SML is just a graphical representation of CAPM.
Slope of the CML is the Sharpe ratio: $\frac{R_m - R_f}{\sigma_m}$	Slope of the SML is the market risk premium: $E(R_{mkt}) - R_f$
Similarities between CML and SML	
Both CML and SML have expected portfolio return on the y-axis.	
The line stretches from the risk-free asset to the market portfolio in both the cases.	

Example

Ricky Ponting invests 10% in the risk-free asset, 40% in a mutual fund which tracks the market and 50% in a high-risk stock with a beta of 2.5. The risk-free rate is 5% and the expected market return is 10%. What is the portfolio beta and expected return?

Solution:

Risk-free asset: w = 0.1 r = 5% β = 0
Mutual fund: w = 0.4 r = 10% β = 1
High-risk stock: w = 0.5 r =? β = 2.5

Portfolio beta = weighted average beta of all assets = 0.1 x 0 + 0.4 x 1 + 0.5 x 2.5 = 1.65

Portfolio return = $r_f + \beta(r_m - r_f)$ = 0.05 + 1.65 [0.1 - 0.05] = 0.1325 = 13.25%

Alternative method:

First determine the return of the high-risk stock using its beta.

Expected return of high-risk stock = 0.05 + 2.5[0.1 - 0.05] = 0.175 = 17.5%

Portfolio return= weighted average return of all assets = 0.1 x 5 + 0.4 x 10 + 0.5 x 17.5 = 13.25%

9. Capital Asset Pricing Model: Applications

The CAPM is important both from a theoretical and practical perspective. In this section, we will look at some of the practical applications of CAPM in capital budgeting, performance appraisal, and security selection.

Estimate of Expected Return

Given an asset's systematic risk, the expected return can be calculated using CAPM.

- To estimate the current price of an asset, we discount future cash flows at the required rate of return calculated using CAPM.
- The required rate of return from the CAPM rate is also used in the capital budgeting process and to determine if a project is economically feasible.

10. Beyond CAPM: Limitations and Extensions of CAPM

Note: there is no explicit learning outcome associated with this section.
In this reading, we saw one return-generating model, the CAPM. But there are more models to estimate the return of an asset.

CAPM is popular because of its simplicity to estimate the expected return. However, there are several theoretical and practical limitations because of its unrealistic assumptions.

Limitations of the CAPM

Theoretical limitations of the CAPM are as follows:

- <u>Single–factor model</u>: Only systematic or market risk is priced. This assumption is restrictive as no other investment characteristic is considered.
- <u>Single-period model</u>: One of the assumptions of the model is that all investors hold assets for a single period, which is practically not true.

Practical limitations of the CAPM are as follows:

- **Market Portfolio:** A market portfolio must comprise all assets, including non-investable assets like Eiffel Tower, human capital, etc.
- **Proxy for a Market Portfolio:** When a true market portfolio comprising all assets cannot be created, a proxy such as S&P500 is used. But different analysts may use different proxies for the same asset based in the country. For example, one analyst may use S&P 500 as the proxy for equity while another may use DAX.
- **Estimation of beta risk:** Beta is an important input for the CAPM model. If not estimated correctly, the expected return will not be accurate as well. Beta is estimated using a long history of returns, which may vary according to the period used. For example, beta calculated using daily returns will be different than a one-year or five-year beta. Similarly, the risk of a company in the past may not represent its current or future risk.
- **CAPM does not predict returns accurately:** Studies have shown that actual returns do not reflect predicted returns.
- **Homogeneity in investor expectations:** CAPM assumes that all investors have the same expectations for securities that result in one optimal risky portfolio, the market portfolio. If investors have different views, then it will result in multiple optimal risky portfolios and SMLs.

Extensions to the CAPM

Other models are considered because of the limitations of CAPM. Of course, these models too come with their own limitations. The models can be broadly categorized into theoretical and practical models.

Theoretical models

In principle, theoretical models are similar to the CAPM but with additional risk factors. One example is the arbitrage pricing theory (APT) which takes the following form:

$\mathbf{E(R_p) = R_F + \lambda_1\beta_{p,1} + \ldots + \lambda_K\beta_{p,k}}$ where k is the number of risk factors, $\boldsymbol{\lambda_1}$ is the risk premium and $\boldsymbol{\beta_{p,k}}$ is the sensitivity of the portfolio to factor k.

The drawback of this model is that it is difficult to identify risk factors and estimate sensitivity to each factor.

Practical Models

The Fama-French three-factor model and four-factor model have been found to predict asset returns better than the CAPM, which considers only beta risk. The three factors included in the Fama-French model are relative size, relative book-to-market value, and beta of the asset. The four-factor model adds one more momentum factor to the three-factor model. These models have limitations too. They cannot be applied to all assets and there is no certainty that these will work in the future.

11. Portfolio Performance Appraisal Measures

Before selecting an investment manager, it is important for investors to understand the performance of a manager and the cost structure involved. In this section, we look at four measures that are commonly used in performance evaluation. These are:

Sharpe Ratio	$\frac{\text{Portfolio risk premium}}{\text{Portolio total risk}} = \frac{R_p - R_f}{\sigma_p}$
M-Squared	$M^2 = \frac{(R_p - R_f)\sigma_m}{\sigma_p} - (R_m - R_f)$
Treynor Ratio	$\frac{\text{Portfolio risk premium}}{\text{beta risk}} = \frac{R_p - R_f}{\beta_p}$
Jensen's Alpha	Actual portfolio return – expected return $= R_p - [R_f + \beta(R_m - R_f)]$

The Sharpe Ratio

Sharpe ratio is the excess return of the portfolio over the risk-free rate divided by the portfolio risk. It is the excess return per unit of risk. The higher the Sharpe ratio the better, all else equal. Sharpe ratio is the slope of the capital allocation line and represents the reward-to-variability ratio. The ex-ante and ex-post Sharpe ratio are given by:

Ex ante SR $= \frac{E(R_p)-R_F}{\sigma_p} \rightarrow$ evaluate the expected risk-adjusted return of the portfolio

Ex post $\widehat{SR} = \frac{\bar{R}_p - \bar{R}_F}{\hat{\sigma}_p}$ → to evaluate historical risk-adjusted returns.

Advantages of Sharpe ratio
- Sharpe ratio can be easily calculated by using readily available market data.
- The Sharpe ratio is easy to interpret.
- The Sharpe ratio is the most widely recognized and used appraisal measure.

Limitations of Sharpe ratio
- The Sharpe ratio uses total risk; not systematic risk.
- The ratio itself is not informative. The Sharpe ratio of one portfolio must be compared with another to see which one is better. For instance, if a portfolio has a Sharpe ratio of 0.7, the number does not convey anything. But if there is another portfolio with a Sharpe ratio of 0.9, then we know it is superior to the one with 0.7.

The Treynor Ratio

Treynor ratio is the excess return of the portfolio over the risk-free rate divided by the systematic risk of the portfolio. The numerator must be positive for meaningful results. It does not work for negative beta assets.

The *ex-ante* and *ex-post* Treynor ratios are provided below.

$$TR = \frac{E(R_p) - R_f}{\beta_p}$$

$$\widehat{TR} = \frac{\bar{R}_p - \bar{R}_f}{\hat{\beta}_p}$$

Limitations of Treynor ratio
- The ratio itself is not informative. When two portfolios are compared, we know which one is superior but we do not know if its performance is better than the market portfolio.
- There is no information about the amount of underperformance or over performance. For instance, assume there are two portfolios A and B with Treynor ratios of 0.6 and 0.7 respectively. Though B is better than A, we have no information as to how much B's performance is better than A.

M-Squared

If M-Squared return (also known as risk adjusted performance measure or RAP) is greater than zero, the manager (portfolio) has positive risk-adjusted return. One way to get a positive M-Squared is when the risk is same as the market but R_P is greater than R_M. Another way is when the return is same as the market but at a lower risk. If M-Squared is zero, then the manager (portfolio) has the same risk-adjusted return as the market. If M-Squared return is negative, the manager (portfolio) has a lower risk-adjusted return than the market.

The *ex-ante* and *ex-post* M^2 are given by:

$$M^2 = \left[E\left(R_p\right) - R_f\right]\frac{\sigma_m}{\sigma_p} + R_f = \text{SR} \times \sigma_m + R_f \text{ (ex ante)}$$

$$\widehat{M^2} = \left(\bar{R}_p - \bar{R}_f\right)\frac{\hat{\sigma}_m}{\hat{\sigma}_p} + R_f = \widehat{\text{SR}} \times \hat{\sigma}_m + R_f \text{ (ex post)}$$

where: σ_m is the standard deviation of the market portfolio and σ_m/σ_p is the portfolio-specific leverage ratio.

The difference between the risk-adjusted performance of the portfolio and the performance of the market is called **M^2 alpha**.

Advantage: M-Squared is in units of the percent return which makes it more intuitive for the interpretation by the user.

Limitation: A limitation of this measure is that it uses total risk and not systematic risk.

Jensen's Alpha

Jensen's alpha is the difference between the actual return on a portfolio and the CAPM calculated, expected or required return. In other words, it is the plot of the excess return of the security on the excess return of the market. The intercept is Jensen's alpha and beta is the slope. Jensen's alpha can be calculated on both ex ante and ex post basis.

Like the Treynor ratio, it is based on systematic risk. Alpha is used to rank different managers and their portfolios. Since Jensen's alpha uses systematic risk, it is theoretically superior to M-Squared.

Interpreting Jensen's Alpha

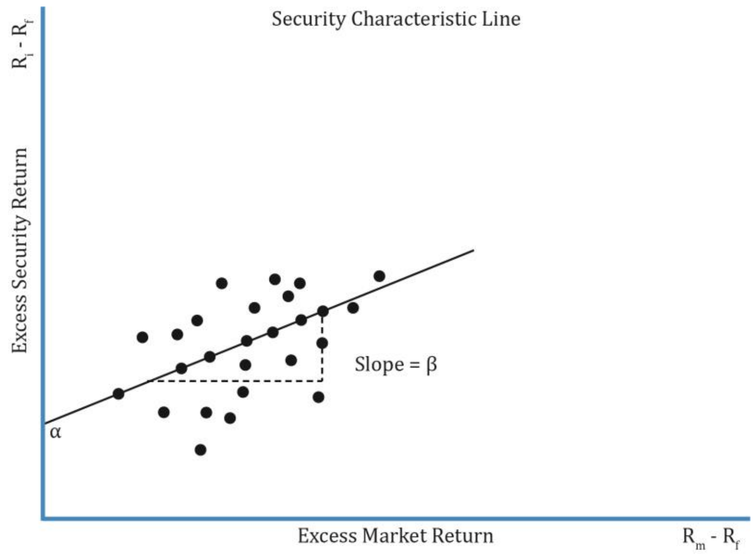
- If Jensen's alpha > 0, the manager (portfolio) has positive risk-adjusted return.
- If Jensen's alpha = 0, then the manager (portfolio) has the same risk-adjusted return as the market.
- If Jensen's alpha < 0, the manager (portfolio) has a lower risk-adjusted return than the market.

Instructor's Note:
Sharpe ratio and M-squared are total risk measures. Use these measures when a portfolio is not fully diversified.

Treynor ratio and Jensen's alpha are based on beta risk and should be used when a portfolio is well diversified.

12. Applications of the CAPM in Portfolio Construction

Security Characteristic Line



The SCL is a plot of the excess return of a security over the risk-free rate on the y-axis, against the excess return of the market on the x-axis. We saw earlier that the SML's intercept on the y-axis is the alpha and the line's slope is its beta. Similarly, the SCL's slope is the security's beta.

The SCL is obtained by regressing excess security return on the excess market return.

SCL: $R_i - R_f = \alpha_i + \beta_i * (R_m - R_f)$

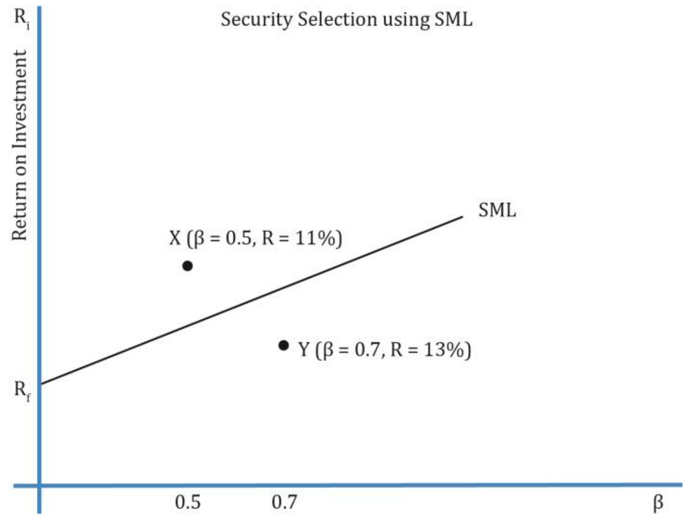
where:
$R_i - R_f$ = excess security return
$R_m - R_f$ = excess market return
α_i = Jensen's alpha or excess return

Security Selection

In CAPM, we assumed that investors have homogeneous expectations and assign the same value to all assets. So, all the investors arrive at the same optimal risky portfolio: the market portfolio. But, in reality, it does not actually happen.

The SML can also be used for security selection. Investors can plot a security's expected return and beta against the SML. The security is undervalued if it plots above the SML. The security is overvalued if it plots below the SML. The security is fairly priced if it plots on the SML.

As you can see in the exhibit below, security X is undervalued as it plots above the SML. At a risk level of $\beta = 0.5$, the return of X is greater than the security that plots on the SML. Similarly, Y must not be bought because the security on SML at a risk level of $\beta = 0.7$ has a higher return than Y.

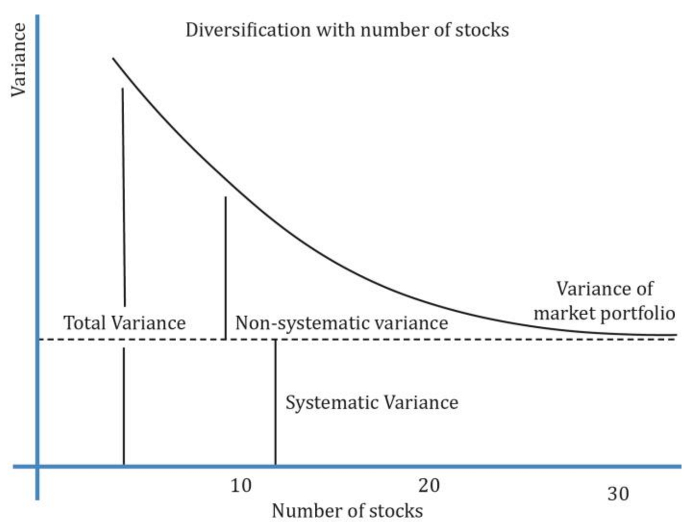


Implications of the CAPM for Portfolio Construction

Theoretically, investors should hold a combination of the risk-free asset and the market portfolio but it is impractical to own the market portfolio as it has a large number of securities. For example, S&P 500 is a good representation of the market as it has 500 stocks. But it can be shown that holding as few as 30 stocks can diversify away the non-systematic risk.

Interpretation of the exhibit below:

- It shows how variance decreases – nonsystematic risk, in particular – as stocks are added to the portfolio.
- Much of the non-systematic risk is diversified away with 30 stocks. As more stocks are added, the non-systematic risk progressively decreases approaching the systematic risk for 30 stocks.
- It is important that these stocks are not correlated with each other and must be randomly selected from different asset classes.



Summary

LO: Describe the implications of combining a risk-free asset with a portfolio of risky assets.

A combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The risk of a portfolio is calculated using the following formula:

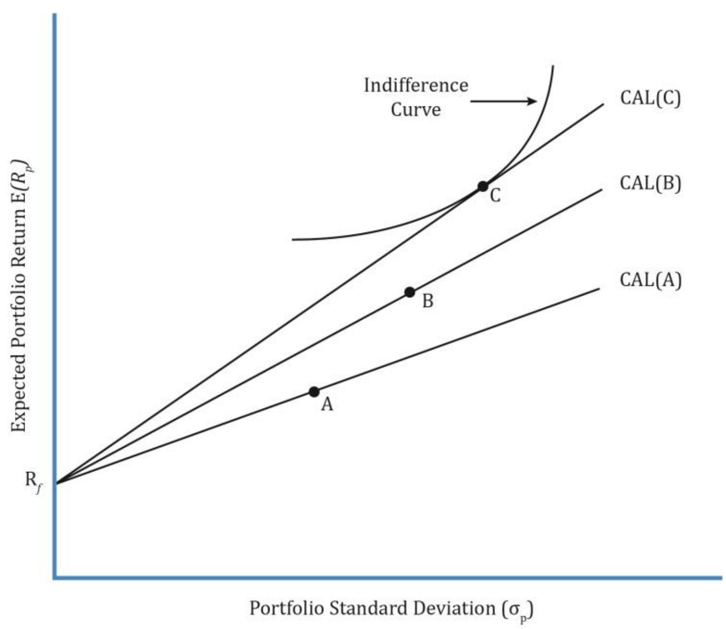
$$\sigma_p = \sqrt{w_1^2 * \sigma_1^2 + w_2^2 * \sigma_2^2 + 2 * w_1 * w_2 * Cov_{1,2}}$$

As a risk-free asset has zero standard deviation and zero correlation of returns with a risky portfolio, it results in the following reduced equation:

$\sigma_p = w_1 * \sigma_1$

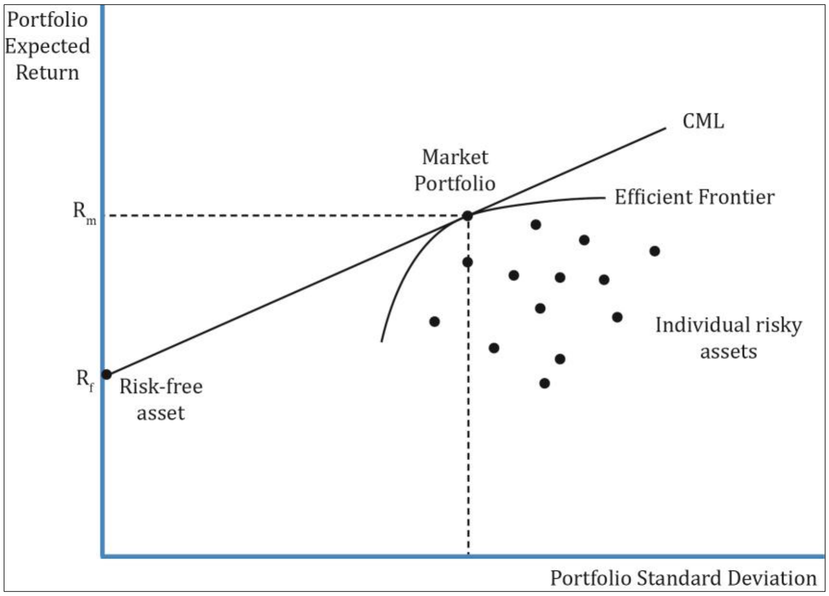
LO: Explain the capital allocation line (CAL) and the capital market line (CML).

Investors have different views of the market, which means the individual risky assets (e.g., securities) they choose to form their portfolio are different. This leads to different optimal risky portfolios, as illustrated in the figure below:



A unique optimal risky portfolio exists for all investors if there is homogeneity of expectations.

If there is one efficient frontier, there will be only one capital allocation line. The point where this capital allocation line is tangential to the efficient frontier is called the market portfolio. This is the optimal risky portfolio when all investors have the same expectations. The CML is a special case of the CAL where the efficient portfolio is the market portfolio.



$$R_p = R_f + \left(\frac{R_m - R_f}{\sigma_m}\right) * \sigma_p$$

LO: Explain systematic and nonsystematic risk, including why an investor should not expect to receive an additional return for bearing nonsystematic risk.

Systematic Risk: It is non-diversifiable or market risk that affects the entire economy and cannot be diversified away. Investors get a return for systematic risk. (Interest rates, inflation rates, natural disaster, political situation, etc.)

Nonsystematic Risk: It is a local risk that affects only a particular asset or industry. There is no compensation for nonsystematic risk as it can be diversified away. (Oil discoveries, non-approval for a drug, new regulations for telecom industry, etc.)

LO: Explain return-generating models (including the market model) and their uses.

Return-generating models provide an estimate of the expected return of a security given certain input parameters called factors.

Multi-factor models include macroeconomic, fundamental, and statistical models.

In a single-factor model, only one factor is considered. A classic single-factor model is the market model which is given by this equation:

$R_i = \alpha_i + \beta R_m + e_i$

LO: Calculate and interpret beta.

Beta is a measure of systematic (or market) risk. It is calculated using the following equation:

$$\beta = \frac{Cov(i, M)}{\sigma_M^2} = \frac{\rho_{iM} * \sigma_i}{\sigma_M}$$

- $\beta > 0$: Return of the asset follows the market trend.
- $\beta < 0$: Return of the asset moves in an opposite direction to the market trend (negatively correlated with the market).
- $\beta = 0$: An asset's return has no correlation with the market. For example, a risk-free asset has a beta of zero.
- $\beta = 1$: Beta of the market is equal to 1. If a stock has a beta of 1, it means it has the same volatility as that of the market.

LO: Explain the capital asset pricing model (CAPM), including its assumptions, and the security market line (SML).

LO: Calculate and interpret the expected return of an asset using the CAPM.

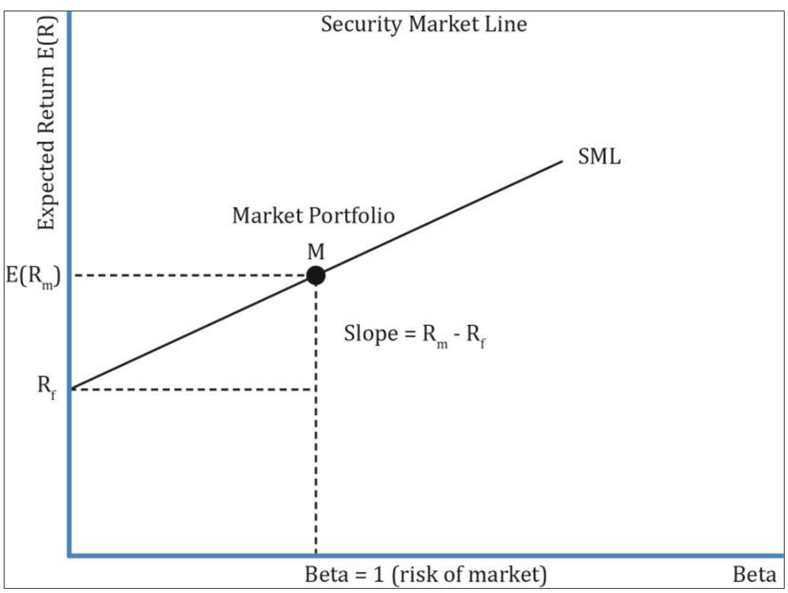
CAPM: $r_e = R_f + \beta[E(R_{mkt}) - R_f]$

The assumptions of CAPM are:

- Investors are risk-averse, utility maximizing, rational individuals.
- Markets are frictionless. There are no transaction costs and no taxes.
- All investors plan for the same, single holding period.
- Investors have homogeneous expectations.
- Investments are infinitely divisible.
- Investors are price takers.

LO: Describe and demonstrate applications of the CAPM and the SML.

The security market line (SML) is a graphical representation of the capital asset pricing model and applies to all securities, whether they are efficient or not. The security is undervalued if it plots above the SML. The security is overvalued if it plots below the SML. The security is fairly priced if it plots on the SML.



Differences between CML and SML	
CML	**SML**
Does not apply to all securities; applies to only efficient portfolios. Only systematic risk is priced. So CAL/CML can only be applied to those portfolios whose total risk is equal to systematic risk.	Applies to any security. It may include inefficient portfolios as well.
X-axis has the standard deviation of the portfolio.	X-axis has beta of the asset or portfolio.
$R_p = R_f + (\frac{R_m - R_f}{\sigma_m})* \sigma_p$ where σ_p is the total risk of the portfolio.	$r_e = R_f + \beta[E(R_{mkt}) - R_f]$ Where β is the systematic risk. SML is just a graphical representation of CAPM.
Slope of the CML is the Sharpe ratio: $\frac{R_m - R_f}{\sigma_m}$	Slope of the SML is the market risk premium: $E(R_{mkt}) - R_f$
Similarities between CML and SML	
Both CML and SML have expected portfolio return on the y-axis.	
The line stretches from the risk-free asset to the market portfolio in both the cases.	

LO: Calculate and interpret the Sharpe ratio, Treynor ratio, M2, and Jensen's alpha.

The four measures commonly used in performance evaluation are:

Sharpe Ratio	$\frac{\text{Portfolio risk premium}}{\text{portolio total risk}} = \frac{R_p - R_f}{\sigma_p}$
M-Squared	$M^2 = \frac{(R_p - R_f)\sigma_m}{\sigma_p} - (R_m - R_f)$
Treynor Ratio	$\frac{\text{Portfolio risk premium}}{\text{beta risk}} = \frac{R_p - R_f}{\beta_p}$
Jensen's Alpha	Actual portfolio return – expected return $= R_p - [R_f + \beta(R_m - R_f)]$

LM04 Basics of Portfolio Planning and Construction

Version 1.0

1. Introduction

This reading addresses the following topics:

- What is an investment policy statement (IPS) and what does it contain?
- The portfolio construction process.
- How is asset allocation done for a client?

2. Portfolio Planning, the IPS and Its Major Components

2.1 The Investment Policy Statement

Portfolio planning can be defined as a program developed in advance of constructing a portfolio that is expected to satisfy the client's investment objectives. The written document governing this process is the investment policy statement (IPS). It defines a plan for investment success given the client's situation and requirements. The IPS should be reviewed on a regular basis.

2.2 Major Components of an IPS

The major components of an IPS are:

- Introduction: Describes the investment objectives, circumstances, and state of client.
- Statement of Purpose: Covers the scope of the IPS.
- Statement of Duties and Responsibilities: Applies to investment manager, client, and other parties involved.
- Procedures: Methodologies to tackle various circumstances and updating the IPS.
- Investment Objectives: Desired rate of return and the amount of risk the client is willing to take.
- Investment Constraints: Liquidity, legal, taxes, time horizon, and other unique constraints.
- Investment Guidelines: Specifies permitted classes, selection of asset classes, use of leverage, asset allocation, and rebalancing, etc.
- Evaluation of performance: Specifies the benchmark portfolio to compare investment results with, the frequency of evaluation, and other related information.
- Appendices: Contains information on specific guidelines like permitted deviations, strategic asset allocation, rebalancing strategies, statement of policy concerning hedging risks such as currency risk and interest rate risk, etc.

3. IPS Risk and Return Objectives

Risk Objectives

A client's risk tolerance is generally expressed qualitatively as high, moderate, or low. Two factors determine the overall risk tolerance: ability and willingness to take risk:

- Ability to take risk is based on wealth, time horizon, and expected income. etc. It is relatively easy to determine. Risk tolerance is usually expressed in both terms: ability

and willingness. For example, a salaried person close to retirement with modest savings has low ability to take risk.
- Willingness to take risk is more subjective and is based on the client's psychology. For example, a client who has recently lost his job may not be willing to take a risk, even though he has the ability to do so, as job loss has a huge psychological impact.

Risk Tolerance

The interactions between willingness to take risk and ability to take risk are shown in the table below:

	Ability to Bear Risk	
Willingness to Take Risk	**Below Average**	**Above Average**
Below Average	Below-average risk tolerance	Resolution needed
Above Average	Resolution needed	Above-average risk tolerance

Source: CFAI

When the ability to take risk and willingness to take risk are consistent, selecting an appropriate level of risk is easy. If the investor's willingness to take risk is high, but his ability to take risk is low, the low ability will dominate and the advisor should select a low risk level. If the investor's willingness to take risk is low, but his ability to take risk is high, the advisor should attempt to educate the investor and clear any misconceptions about investment risk. If the investor is still not willing to take more risk, the advisor should select a low risk level.

For some clients, the risk objective might be defined quantitatively. Quantitative risk objectives can be expressed in absolute or relative terms.

- <u>Absolute risk objective example</u>: Portfolio should not suffer more than a 5% loss in any 12-month period. Practically, this could be stated as: with 95% probability, portfolio should not lose more than 5% value in any 12-month period. Absolute risk measures are not related to market performance. They are expressed in terms of standard deviation, variance, or value at risk.
- <u>Relative risk objective example</u>: The risk objective is expressed relative to a benchmark. For example, return should be within 4% of the S&P 500 index return.

Example

Your client has a portfolio worth 10 million. He cannot handle losing more than 1 million over the next 12 months. Is this an absolute or relative risk objective? How can this be stated in practical terms?

Solution:

This is an absolute risk objective. In practical terms, it can be stated as: with 95% probability, portfolio should not lose more than 10% in the next 12 months.

Example

Another client specifies a risk objective of achieving returns within 4% of the BSE 100. Is this an absolute or relative risk objective? Identify a measure for quantifying the risk objective.

Solution:

This is a relative risk objective as it is relative to BSE 100 (market) performance. A measure for tracking a relative risk objective is tracking the risk.

3.1 Return Objectives

Return objectives can be stated on an absolute or a relative basis.

- Absolute: Absolute return is the return a portfolio must achieve over a certain period of time. For example, a client wants to achieve a return of 9% or inflation-adjusted (real) return of 2%. The objective is to deliver a positive return over time, irrespective of how good or bad the market performance is. No index or benchmark is used to measure the performance. Many strategies may be employed to generate absolute return, the success of which depends on the skills of the manager.
- Relative: A relative return objective will be stated relative to a benchmark. Examples: return 3% greater than 12-month LIBOR or return equal to the S&P500 index return.

The return objective can be stated before or after fees, and pre- or post-tax. The fee structure must be clear and understood by both the parties, i.e., the investment manager and the client. If there is a required return that must be met for the client to meet a specific goal, such as down payment of $100,000 for a house next year or $20,000 for college education for a child next year, then it must be mentioned.

Stated risk and return must be compatible. For example, it would be unrealistic to expect a very high return with low risk tolerance.

Example

Your client is 35 and wishes to retire in 30 years. His salary meets current and expected future expenses. He has 100,000 in savings of which he wants to put aside 10,000 as an emergency fund to be held in cash. You estimate that 300,000 in today's money will be sufficient to fund your client's retirement income needs. Expected inflation is 2% over the next 30 years. How much money must your client have in nominal terms to fund his retirement? What is the required return objective?

Solution:

First, let us calculate how much money the client must have in nominal terms after 30 years, at his retirement. You can solve it two ways:

Using the formula: 300,000 to grow at 2% for 30 years = $300{,}000 * (1.02)^{30}$ = 543,408

Using a financial calculator: N = 30, I/Y = 2, PV = -300,000, PMT = 0, CPT FV. FV = 543,408

To calculate the return objective, we have the following data.

The client is investing 90,000 now after keeping aside 10,000 for emergency in cash. This 90,000 must grow to 543,408.47 in 30 years. To compute the interest rate, we enter the following values in the calculator:

FV = -543,408.47, PV = 90,000, N = 30, PMT = 0, CPT I/Y. Interest rate = 6.18%

Note: If the client is saving a particular amount every year, then key in that number for PMT.

4. IPS Constraints

The five major investment constraints are:

- Liquidity requirements**:**
  - The ability to convert invested assets into cash without suffering significant price erosion.
  - Cash requirement varies from client to client and may require a certain portion of assets to be invested in highly liquid investments.
- Legal and Regulatory**:**
  - Restrictions on investments and percentage allocation in certain assets for investors, like insurance firms, trusts, etc.
- Time Horizon**:**
  - The longer the time horizon, the greater is the ability to take risk and the lower are the liquidity needs in the portfolio.
- Tax Concerns**:**
  - Investors tax status, jurisdiction of investments, and the tax treatment of various types of investment accounts should be considered.
- Unique Circumstances**:**
  - Factors influenced by religion, ethical preferences, government policies, or investor circumstances (including beliefs and values).

Instructor's Note:

You can use the acronym 'LLTTU' to remember these five constraints.

IPS may also include policy regarding **responsible investing** which takes into account environmental, social, and governance (ESG) factors. The six main ESG investment approaches are:

- Negative screening: Excluding certain sectors or companies or practices from a fund or portfolio on the basis of specific ESG criteria.
- Positive screening: Including certain sectors, companies, or practices in a fund or portfolio on the basis of specific ESG criteria.
- ESG integration: Refers to the practice of including *material* ESG factors in the investment process.

- Thematic investing: This strategy picks investments based on a theme or single factor, such as energy efficiency or climate change.
- Engagement/active ownership: This strategy involves achieving targeted social or environmental objectives along with measurable financial returns by using shareholder power to influence corporate behavior.
- Impact investing: Investments made with the intention to generate positive, measurable social and environmental impact alongside a financial return.

These ESG investment approaches can impact the portfolio manager's investment universe and may also require the manager to put in place a process to systematically incorporate ESG factors into the portfolio.

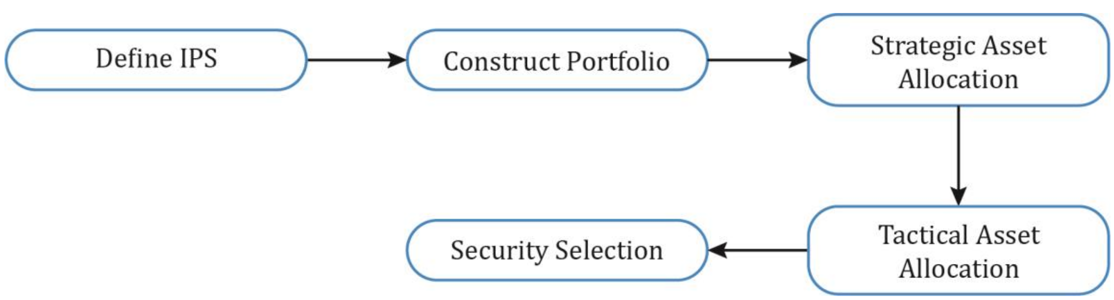
5. Gathering Client Information

It is important for portfolio managers and investment advisers to know their clients. They must find out all the facts about the client at the start of the relationship. This includes collecting information about the client's current circumstances, spending requirements, return objectives, goals, etc. Some of the important data gathered include:

- Family situation: Married or not, if the spouse works, any additional dependents, number of children and their education plans, etc.
- Employment situation: Client's salary, sources of income, industry the client is working in, stability of job, etc.
- Financial information: Level of savings, other investments such as real estate, etc. Adequate information on financial position will help in evaluating the client's risk tolerance.

6. Portfolio Construction and Capital Market Expectations

We have defined the IPS with return and risk objectives, and five constraints. Now, using the points in the IPS as a guideline, we need to construct the portfolio.



Portfolio construction consists of three steps:

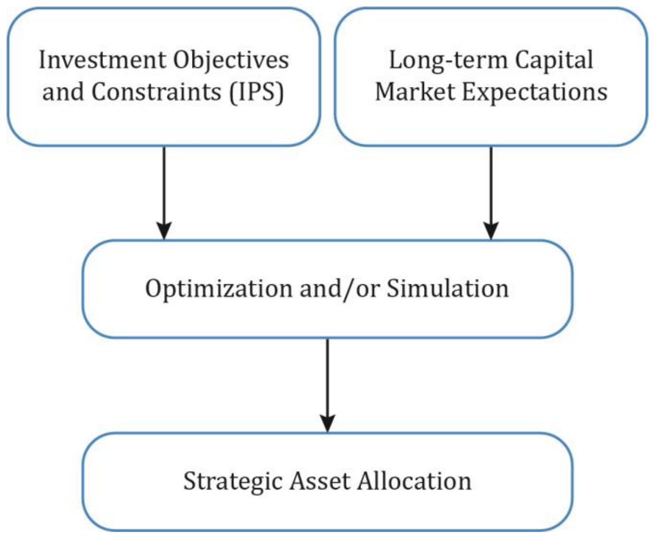
- Strategic asset allocation
- Tactical asset allocation
- Security selection

6.1 Capital Market Expectations

Capital market expectations are the investor's expectations about the return and risk of various asset classes. Capital market expectations include the return for each asset class an investor may invest in (e.g., stock market, bond market, alternative investments, real estate, etc.), the standard deviation of returns for each asset class (risk), and the correlation between the asset classes.

7. The Strategic Asset Allocation

The long-term capital market expectations and investor's risk-return objectives are combined into a strategic asset allocation. This is accomplished through optimization and/or simulation on computer systems.

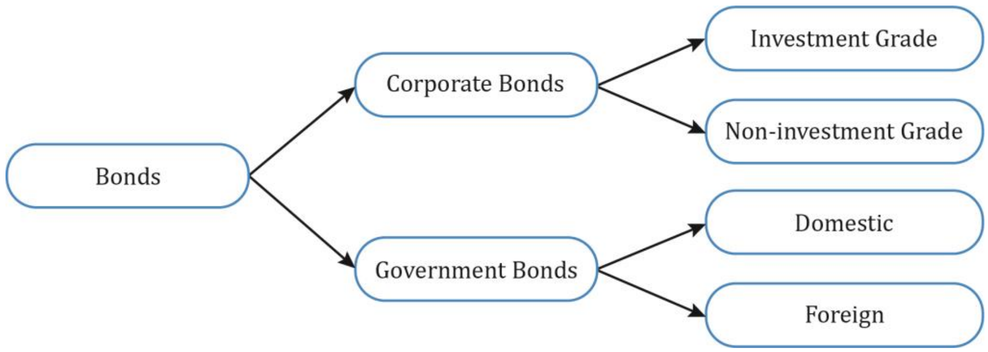


Strategic asset allocation is a strategy to allot a certain percentage of the portfolio, each to different IPS-permissible asset classes, in order to achieve the client's long-term goals. Using this method, the portfolio manager decides how much of the client's money should be invested in equities, bonds, or any other asset class to meet the client's long-term goals. Strategic asset allocation is important because:

- Most of a portfolio's returns come from its systematic risk as nonsystematic risk is diversified away.
- The returns of assets in an asset class reflect exposure to certain systematic factors. This information can be used to select asset classes that match an investor's risk and return objectives.

How are asset classes defined?

The classification of asset classes is somewhat subjective. Furthermore, an asset class can be divided into sub-asset classes as illustrated below.



Criteria to define asset classes:

- All assets in an asset class must be homogeneous, and not correlated to other asset classes.
- Correlations of assets with an asset class should be high.
- Risk and return expectations of assets within an asset class must be similar.
- All the asset classes combined should account for the universe of all investable assets.

When defining the SAA, it is important to consider the asset class correlation matrix. When the correlation between asset classes is low, the diversification benefit will be high. This concept has been discussed in detail in earlier readings.

Example

Given the matrix below, identify which asset class is most sharply distinguished from equities.

Historical correlation (May 31, 2005 to April 30, 2009)

	Equities	Fixed Income	Hedge	Real Estate	Private Equity	Commodities	Currencies
Equities	1.00						
Fixed Income	-0.35	1.00					
Hedge	0.64	-0.35	1.00				
Real Estate	0.88	-0.21	0.58	1.00			
Private Equity	0.88	-0.30	0.65	0.92	1.00		
Commodities	0.38	-0.37	0.60	0.29	0.45	1.00	
Currencies	0.18	0.16	0.19	0.16	0.16	0.26	1.00

Source: FT Alphaville

Solution:

The question asks us to identify the asset class with the lowest correlation with equities. As you can see from the table, fixed income has the lowest correlation with equities while real estate and private equity have the highest correlation with equities.

Once the asset allocation is done, it is possible for this asset allocation to drift from the target allocation with time. For example, let us assume the target asset allocation is 60 percent in stocks and 40 percent in bonds. If equities do well the following year, the asset allocation drifts to 90 percent in equities and 10 percent in bonds. This calls for rebalancing the portfolio as the drift is substantial. By rebalancing, we mean sell equities and buy bonds to bring the portfolio back to the target asset allocation. The amount of allowable drift and rebalancing policy should be defined in the IPS appendix. This material will be covered in detail at Level III.

8. Steps toward an Actual Portfolio and Alternative Portfolio Organizing Principles

Portfolio construction involves the following steps:

1. **Define IPS:**
   a. Capture the investor's requirements and constraints.
2. **Determine the strategic asset allocation:**
   a. Define the investable asset classes for the portfolio and gather historical data on their risk, return, and correlation.
   b. Combine the IPS and the risk/return profile of various portfolios, derived from the above step, to decide on a strategic asset allocation for the portfolio. Until this step, investment decisions are entirely passive, i.e., returns are primarily generated by investing in asset class indexes.
   c. The SAA is a means of providing the investor with exposure to the systematic risks of asset classes in proportions that meet the risk and return objectives. It drives a significant percentage of the overall return.
3. **Tactical asset allocation:**
   a. This is the first step of active management where asset classes are selected.
   b. Determine whether there are any short-term opportunities that warrant a deviation from the strategic asset allocation.
   c. The weights of asset classes are altered from the strategic allocation weights.
   d. For example, a top-down analysis shows that given the economic cycle, commodities may outperform. Based on this premise, you alter the weight for the commodity asset class.
4. **Security selection:**
   a. This is second step of active management, where particular securities are selected.
   b. Identify the relatively strong securities within the favored asset class.
   c. Increase the weights of these securities from the weights used in index construction, to outperform the benchmark.
   d. For example, in your analysis you decide to go overweight on the base metals securities.

Some additional terms you should know:

- <u>Risk budgeting</u>: The process of deciding on the amount of risk to assume in a portfolio (the overall risk budget) and subdividing that risk over the sources of investment return (e.g., strategic asset allocation, tactical asset allocation, and security selection).

- Passive versus active investing: Passive investing is a strategy in which investors invest based on a pre-defined benchmark. One example would be an investment in a fund that tracks the S&P 500. Active investing is a strategy to identify (buy) underpriced and (sell) overpriced stocks. The objective is to earn a return higher than the benchmark.
- Rebalancing policy**:** The process of restoring a portfolio's original exposures to systematic risk factors is defined in the rebalancing policy.

8.1 New Developments in Portfolio Management

Two new developments in portfolio management are:

- Increasing use of ETFs in combination with robo-advice by retail investors: ETFs provide a fast, inexpensive, and liquid exposure to asset classes. Robo-advice has also further reduced the costs for retail investors to create well-diversified portfolios.
- Use of risk parity investing approach: Another new development with respect to portfolio planning and construction is the use of risk parity investing, whereby, asset classes are weighted according to risk contribution.

9. ESG Considerations in Portfolio Planning and Construction

ESG implementation approaches require a set of instructions for investment managers regarding selection of securities, the exercise of shareholder rights and the selection of investment strategies.

ESG implementation approaches affects both strategic asset allocation and the portfolio construction process. They may have a negative impact on expected risk and return of a portfolio as it may limit the manager's investment universe and the manner in which investment management firms operate. Nonetheless, ESG investing continues to see strong adoptions. Responsible investing proponents argue that the potential improvements in governance and avoidance of material risks will enhance returns. Academic research on the impact of ESG factors on portfolio returns remains mixed.

Summary

LO: Describe the reasons for a written investment policy statement (IPS).

The IPS is the starting point of the portfolio management process. Before constructing a portfolio or choosing assets for a client, it is important to understand the client's objectives. How much risk is he willing to take, how much can he actually take, how much return does he expect from the portfolio, what are his current circumstances? It defines a plan for investment success given the client's situation and requirements.

LO: Describe the major components of an IPS.

The major components of an IPS are:

- Introduction: Describes the investment objectives, circumstances, and state of the client.
- Statement of Purpose: Covers the scope of the IPS.
- Statement of Duties and Responsibilities: Applies to investment manager, client, and other parties involved.
- Procedures: Methodologies to tackle various circumstances and updating the IPS.
- Investment Objectives: The desired rate of return and the amount of risk the client is willing to take.
- Investment Constraints: Liquidity, legal, taxes, time horizon, and other unique constraints.
- Investment Guidelines: Specify permitted asset classes, selection of asset classes, use of leverage, asset allocation, and rebalancing etc.
- Evaluation of performance: Specifies the benchmark portfolio to compare investment results with, the frequency of evaluation, and other related information.
- Appendices: Contain information on specific guidelines like permitted deviations, strategic asset allocation, rebalancing strategies, etc.

LO: Describe risk and return objectives and how they may be developed for a client.

Risk objective might be defined quantitatively. Quantitative risk objectives can be expressed in absolute or relative terms.

- Absolute risk objective example: The portfolio should not suffer more than a 5% loss in any 12-month period. Practically, this could be stated as: with 95% probability, the portfolio should not lose more than 5% value in any 12-month period. The absolute risk measures are not related to market performance. They are expressed in terms of standard deviation, variance, or value at risk.
- Relative risk objective example: Return should be within 4% of the S&P 500 index return. The risk objective is expressed relative to a benchmark.

Return objectives can be stated on an absolute or a relative basis.

- Absolute: Absolute return is the return a portfolio must achieve over a certain period of time.

- Relative: A relative return objective will be stated relative to a benchmark.

LO: Explain the difference between the willingness and the ability (capacity) to take risks in analyzing an investor’s financial risk tolerance.

Two factors determine the overall risk tolerance: ability and willingness to take risk:

- Ability to take risk is based on wealth, time horizon, expected income, etc. It is relatively easy to determine. For example, a salaried person close to retirement, with modest savings, has a low ability to take risks.
- Willingness to take risk is more subjective and is based on the client’s psychology. For example, a client who has recently lost his job may not be willing to take risks, even though he has the ability to do so, as job loss has a huge psychological impact.

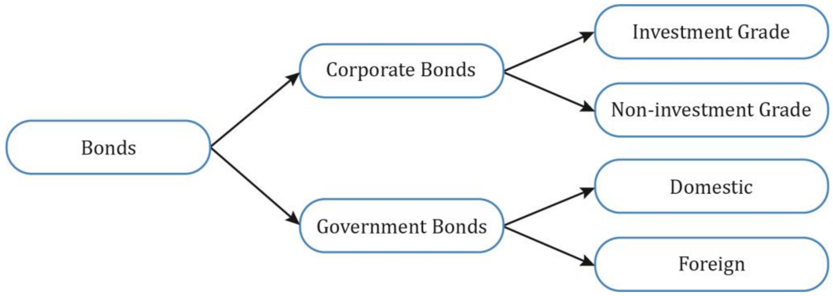
LO: Describe the investment constraints of liquidity, time horizon, tax concerns, legal, and regulatory factors, and unique circumstances and their implications for the choice of portfolio assets.

The five major investment constraints are:

- Time Horizon:
  - The longer the time horizon, the greater is the ability to take risk and the lower are the liquidity needs in the portfolio.
- Tax Concerns:
  - The investor’s tax status, jurisdiction of investments, and tax treatment of various types of investment accounts should be considered.
- Liquidity requirements:
  - The ability to convert invested assets into cash without suffering significant price erosion.
  - Cash requirement varies from client to client and may require a certain portion of the assets to be invested in highly liquid investments.
- Legal and Regulatory:
  - Restrictions on investments and percentage allocation in certain assets for investors like insurance firms, trusts, etc.
- Unique Circumstances:
  - Factors influenced by religion, ethical preferences, government policies, or investor circumstances.

LO: Explain the specification of asset classes in relation to asset allocation.

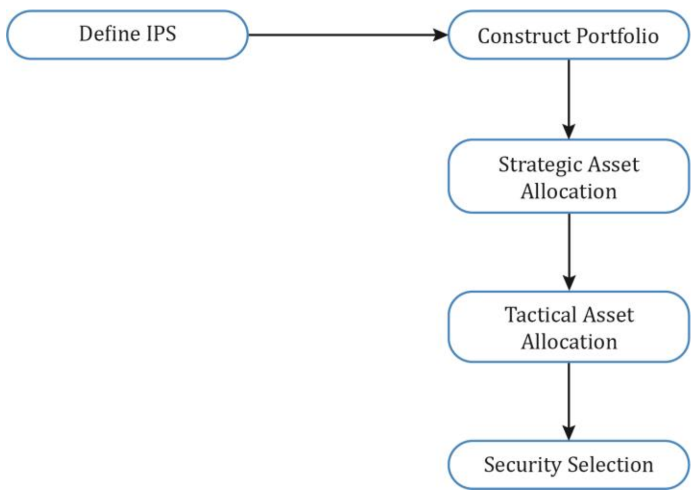
The classification of asset classes is somewhat subjective. Furthermore, an asset class can be divided into sub-asset classes as illustrated below.



- All assets in an asset class must be homogeneous, and not correlated to other asset classes.
- Correlations of assets with an asset class should be high.
- Risk and return expectations of assets within an asset class must be similar.
- All the asset classes combined should account for the universe of all investable assets.

LO: Discuss the principles of portfolio construction and the role of asset allocation in relation to the IPS.

Using the points in the IPS as a guideline, a portfolio is constructed.



<u>Strategic asset allocation</u> is a strategy to allocate a certain percentage of the portfolio to each of the different asset classes in order to achieve the client's long-term goals. Using this method, the portfolio manager decides how much of the client's money should be invested in equities, bonds, or any other asset class to meet the client's long-term goals.

<u>Tactical asset allocation</u> is an active investment strategy that attempts to profit from short-term mispricing. The strategy involves deviating from the intended weights of asset classes in the short-term.

<u>Security selection</u> is the process of determining which securities to include in a portfolio so that the portfolio generates a return higher than the benchmark.

LO: Describe how environmental, social, and governance (ESG) considerations may be integrated into portfolio planning and construction.

IPS may also include policy regarding responsible investing which takes into account environmental, social, and governance (ESG) factors. The six main ESG investment approaches are:

- Negative screening
- Positive screening
- ESG integration
- Thematic investing
- Engagement/active ownership
- Impact investing

The ESG implementation approaches may have a negative impact on expected risk and return of a portfolio as it may limit the manager's investment universe and the manner in which investment management firms operate. Nonetheless, ESG investing continues to see strong adoptions. Responsible investing proponents argue that the potential improvements in governance and avoidance of material risks will enhance returns.

LM05 The Behavioral Biases of Individuals

Ver 1.0

1. Introduction

Traditional finance assumes that individuals act rationally by considering all available information in their decision-making process. This leads to optimal outcomes and efficient markets.

Behavioral finance challenges these assumptions by incorporating research on how individuals and markets actually behave. It assumes that people act "normal" rather than rational. It recognizes that people have behavioral biases that impact financial decision making. Behavioral biases can cause decisions to differ from the "rational" decisions of traditional finance.

This reading is organized as follows:

- Section 2 broadly characterizes behavioral biases into cognitive errors and emotional biases.
- Section 3 explains cognitive errors, the consequences of each bias, detection of the bias, and guidance for overcoming these biases.
- Section 4 describes emotional biases, the consequences of each bias, detection of the bias, and guidance for overcoming these biases.
- Section 5 discusess how behavioral finance influences market behaviour and the occurance of market anomalies

2. Behavioral Bias Categories

Behavioral biases can be either cognitive errors or emotional biases.

Cognitive errors:

- Stem from statistical, information-processing, or memory errors.
- Can occur as a result of faulty reasoning.
- Are more easily corrected than emotional biases.
- Can be corrected through better information, education, and advice.

Emotional biases:

- Stem from impulse or intuition.
- Are influenced by feelings and emotion.
- Are spontaneous.
- Are less easy to correct as compared to cognitive errors.
- Can be recognized and "adapted" to.

3. Cognitive Errors

There are two categories of cognitive biases: 1) belief perseverance biases and 2) information-processing biases.

Instructor's Note: A summary of the nine cognitive errors is provided below. They are covered in detail in the subsequent sections. Memorize this summary.

Belief perseverance biases

- Conservatism bias: Maintain prior views by inadequately incorporating new information.
- Confirmation bias: Look for and notice what confirms their beliefs.
- Representativeness bias: Classify new information based on past experiences.
- Illusion of control bias: False belief that we can influence or control outcomes.
- Hindsight bias: See past events as having been predictable.

Information-processing biases

- Anchoring & adjustment bias: Incorrect use of psychological heuristics.
- Mental accounting bias: Treat one sum of money different from other.
- Framing bias: Answer question differently based on how it is asked.
- Availability bias: Heuristic approach influenced by how easily outcome comes to mind.

3.1 Belief Perseverance Biases

Belief perseverance is the tendency to cling to one's previously held beliefs irrationally or illogically. Belief perseverance biases are closely related to cognitive dissonance. Cognitive dissonance is the mental discomfort that occurs when new information conflicts with old beliefs or cognitions.

The belief perseverance biases discussed in this reading are: conservatism, confirmation, representativeness, illusion of control, and hindsight.

Conservatism Bias

Individuals maintain prior views by inadequately incorporating new information. From the traditional finance perspective, this can be described as the failure to update probabilities using Bayes' formula.

This bias has aspects of both statistical and information-processing errors. It causes individuals to overweight initial beliefs about probabilities and outcomes; and under-react to new information.

Consequences of conservatism bias:

- Maintain or be slow to update a view or a forecast, even when new information is available.
- Opt to maintain a prior belief rather than deal with the mental stress of updating beliefs given complex data. This relates to an underlying difficulty in processing new information.

- As a consequence of conservatism, an investor may hold a security longer than a rational decision maker.

Detecting and overcoming conservatism bias:

- Recognize that bias exists.
- Ask questions when presented with new information: "How does this information change my forecast?" "What is the impact of this information?"
- Updating of beliefs is inversely correlated with the effort involved. The more effort it takes to change a prior view, or the more effort it takes to process new information, the more likely the individual will ignore it.
- Should conduct careful analysis incorporating the new information.
- When it is difficult to interpret new information, seek advice from experts.

Confirmation Bias

Confirmation bias occurs when individuals look for and notice what confirms their beliefs and ignore what contradicts their beliefs.

Embedded Example

A client insists on continuing to hold stock, even when the adviser recommends otherwise because the client's follow-up research seeks only information that confirms his belief that the investment is still a good value.

Consequences of confirmation bias:

- Consider only positive information about an existing investment and ignore negative information.
- Develop screening criteria and ignore information that refutes the screening criteria' validity or supports other screening criteria.
- Under-diversify portfolios, leading to excessive exposure to risk.
- Hold a disproportionate amount of their investment assets in their employing company's stock. Because of the confirmation bias, the FMP(Financial market participant) is convinced of the company's favorable prospects and ignores unfavorable information.

Detecting and overcoming confirmation bias:

- Seek information that challenges your (FMPs) beliefs.
- Get corroborating support on an investment decision. If you are a technical analyst and you believe that the stock will go up. Before giving investment advice, it is advisable to get supporting information from (say) your colleague, a fundamental analyst. Because your colleague may have a completely different valuation of the stock. It is essential to

make the extra effort to gather complete information, positive and negative, for better decisions.

Representativeness Bias

Representativeness bias is when people classify new information based on past experiences and classifications. The two types of representativeness bias are ‘base-rate neglect’ and ‘sample-size neglect’.

In base-rate neglect the base rate, or probability of the categorization is not adequately considered. For example, categorizing Company ABC as a “growth stock” without appropriate due diligence. FMPs rely on stereotypes when making investment decisions without adequately incorporating the base probability of the stereotype occurring. So, the classification of ABC is based on FMPs stereotypes about growth companies, but it ignores the base probability that the company may not be a growth company.

In sample-size neglect FMPs incorrectly assume that small sample sizes are *representative* of populations (or “real” data). In the investment context, sample size neglect can be seen when a few data points are naïvely extrapolated as being representative of a long-term trend. For example, an investor who puts too much emphasis on short-term results when considering a potential investment.

Consequences of representativeness bias:

- Adopt a view or a forecast based almost exclusively on new information or a small sample.
- Update beliefs using simple classifications rather than deal with the mental stress of updating beliefs given complex data.

Detecting and overcoming representativeness bias:

- Be aware of statistical mistakes you may be committing.
- Continually ask yourselves, “Are you overlooking the reality of the investment situation?”

Illusion of Control Bias

Illusion of control bias occurs when individuals incorrectly believe that they can control or influence outcomes when they cannot.

Consequences of illusion of control:

- Trade more than is prudent. For example, day traders believe that they have “control” over their investments’ returns. This view leads to excessive trading, which may lead to lower realized returns.

- Inadequate diversification. An investor prefers to invest in a company where he works because he feels he controls its future. This leads to concentrated positions.

Detecting and overcoming illusion of control bias:

- Recognize that investing is a probabilistic activity.
- Global capitalism is complex; even the most powerful investors have little control over the outcomes of their investments. For example, AIG did not have control over what happened in 2008-2009.
- Seek contrary viewpoints.
- Maintain records of your transactions, the rationale behind each trade, the attributes that you have determined to be in favor of the investment's success. This will make you realize that you do not control the outcome of the investment.

Hindsight Bias

Hindsight bias is a bias where people may see past events as having been predictable and reasonable to expect. In hindsight, poorly reasoned decisions with positive results may be described as brilliant tactical moves, and poor results of well-reasoned decisions may be described as avoidable mistakes. Also, people will remember their own predictions of the future as more accurate than they actually were because they are biased by the knowledge of what has actually happened.

Consequences of hindsight bias:

- Overestimate the degree to which they predicted an investment outcome, which can lead to a false sense of confidence.
- Unfairly assess a money manager's or security's performance. For example, a money manager might have performed well in the past, given his value-investing strategy. But in the present year, growth stocks gave better returns. As a result, investors with the hindsight bias will believe that growth managers show superior performance to value managers because of what has taken place in securities markets. Performance is compared against what has happened as opposed to expectations.

Detecting and overcoming the bias:

- Recognize the bias.
- Ask such questions as, "Am I re-writing history or being honest with myself about the mistakes I made?"
- Carefully record and examine investment decisions, both good and bad.

3.2 Processing Errors

The second category of cognitive error is information-processing biases. These biases result in information being processed and used illogically or irrationally. In contrast to belief perseverance biases, these are less related to errors of memory and more to do with how information is processed. The processing errors discussed in this reading are anchoring and adjustment, mental accounting, framing, and availability.

Anchoring and Adjustment Bias

Anchoring and adjustment bias is an information-processing bias in which the use of a psychological heuristic influences the way people estimate probabilities. People set an anchor that influences their decisions; they adjust the anchor up or down to reflect subsequent information. Irrespective of how the initial anchor was chosen, people often fail to adjust their anchors properly, and as a result produce biased approximations.

This bias is closely related to the conservatism bias. In the conservatism bias, people place undue weight on past information compared to new information. In anchoring and adjustment, people place undue weight on an "anchored" value.

Consequences of anchoring and adjustment bias:

- FMPs may stick too closely to their original estimates when new information is learned. For example, an FMP originally bought a technology stock for $25, and the stock kept doing well until the company started experiencing difficulties during the year. As the stock price dived, the FMP did not adequately adjust the $25 given the difficulties. He remained "anchored" to the $25 he had paid and refused to sell the stock. This mindset is not limited to downside adjustments; the same can happen when companies have upside surprises.

Detecting and overcoming anchoring and adjustment bias:

- Consciously ask questions that may reveal an anchoring and adjustment bias. For example, "Am I holding onto this stock based on rational analysis, or am I trying to attain a price that I am anchored to, such as the purchase price or a high-water mark?"
- Recognize that past prices and market levels are not an indication of what will happen in the future. They should not significantly influence buy and sell decisions.

Mental Accounting Bias

Mental accounting bias is a bias in which people treat one sum of money differently from another sum based on which mental account (layer) the money is assigned to.

An investor may create three layers of money – the first layer includes conservative investments, the second layer has moderate risk investments, and the third layer has high-risk assets. Investors assign investments into discrete "buckets," i.e., several non-fungible mental accounts. This method is problematic and contradicts rational economic thought because money is inherently fungible (interchangeable).

Consequences of mental accounting bias:

- Money is placed in "buckets" or "layers" without considering correlations among assets. FMPs neglect opportunities to reduce risk by combining assets with low correlations.
- Irrationally distinguish between returns derived from income and from capital appreciation. Some FMPs are more comfortable spending the income generated but try to preserve the principal. As a result, they may overweight income generating assets, which may result in a lower total return.
- Irrationaly distinguish between investment principal and investment returns. Some FMPs invest the original principal rationally but are willing to take a very high risk with the investment returns. In the casino sector, "playing with house money" is a typical term used for this situation.

Detecting and overcoming mental accounting bias:

- Recognize the drawbacks of putting money in different buckets.
- Combine all assets onto one spreadsheet to see the true asset allocation of various mental account layers.
- Focus on total return instead of dividing return into income and capital appreciation components. Say you buy a stock that is increasing in value but does not pay dividends. You can always sell a part of your stock and create an income stream.

Framing Bias

Framing bias occurs when an individual answers a question differently, depending on how it is asked (framed).

Embedded Example

A situation may be presented within a gain context (one in four start-up companies succeed) or within a loss context (three out of four start-ups fail).

Given the first frame, an FMP may adopt a positive outlook and make venture capital investments. Given the second frame, the FMP may avoid making the investment.

Consequences of framing bias:

- FMPs' willingness to accept risk can be influenced by how situations are presented or framed. This can result in misidentifying risk tolerances. Because of how questions are framed, FMPs may become more risk-averse when presented with a gain frame of reference and more risk-seeking when presented with a loss frame of reference.
- Choose suboptimal investments based on how information about the specific investments is framed.
- Have a narrow frame when evaluating an investment. Focus on short-term price fluctuations may result in excessive trading.

Detecting and overcoming framing bias:

- Ask such questions as, "Is the decision the result of focusing on a net gain or net loss position?"
- Try to be neutral and open-minded when evaluating investments.

Availability Bias

Availability bias is when people take a heuristic (sometimes called a rule of thumb or a mental shortcut) approach to estimating the probability of an outcome based on how easily the outcome comes to mind. Easily recalled and understandable outcomes are perceived as more likely than harder to recall ones.

The four sources of the bias most applicable to FMPs are retrievability, categorization, narrow range of experience, and resonance.

Retrievability: If an answer or idea comes to mind more quickly than others, it will likely be chosen as correct even if it is not. For example, a study was done in which subjects listened to a list of 50 names and were then asked to judge if the list contained more men or women. The list contained 30 generic women names, but the 20 men were all famous. As availability theory would predict, most of the subjects concluded that the list contained more men than women because they immediately recalled the famous men's names.

Categorization: When solving problems, people gather information from what they recognize as relevant search sets. If it is difficult to come up with a search set, the estimated probability may be biased. For example, in the US, baseball is a popular sport, but soccer is not. If an American is asked to list famous baseball players and famous soccer players, the list of soccer players will be relatively short. An American may incorrectly conclude that there are more famous baseball players than soccer players. Because in their categorization, there are more baseball players than soccer players.

Narrow Range of Experience: This bias occurs when a person with a narrow range of experience uses too narrow a frame of reference based upon that experience when he makes an estimate. For example, assume that an investor advisor has only been in the market for three years. When he makes conclusions or decisions, his availability bias will be based on three years of experience. He would not have seen the great stock market crashes, the booms/bursts of the past.

Resonance: The conclusions or decisions people take are often biased by how closely a situation resonates with their personal experience. For example, jazz music lovers are likely to overestimate how many people listen to jazz music. On the other hand, people who dislike jazz music are likely to overestimate the number of people who dislike jazz music.

Consequences of availability bias:

- Choosing an investment, investment adviser, or mutual fund based on advertising rather than appropriate analysis. For instance, when choosing a mutual fund to invest with, many people will choose the fund that does extensive advertising because its name will easily come to mind. Choices based on advertising are consistent with retrievability as a source of availability bias.
- FMPs limit their investment opportunity set.
- Fail to diversify.
- Fail to achieve an appropriate asset allocation.

Detecting and overcoming availability bias:

- Recognize the bias. It is a human tendency to overemphasize the most recent financial events because of easy recall.
- Follow a disciplined approach to investing. To develop an appropriate investment policy strategy, carefully research, and analyze investment decisions and focus on the long-term.
- Ask such questions: "How did I hear about the stocks?" Was it on Bloomberg, or saw them on CNBC, or read a research report?" and "Am I buying or selling investments because of a recent market event without doing a thorough analysis?"
- Recognize that we forget events that happened a few years ago.

4. Emotional Biases

Instructor's Note: A summary of the six types of emotional biases is presented below. They are covered in detail in the subsequent sections. Memorize this summary. Use the acronym 'LOSSER' to remember this list.

- Loss-aversion bias: Prefer avoiding losses over achieving gains.
- Overconfidence bias: Unwarranted faith in one's abilities.
- Self-control bias: Fail to act in pursuit of long-term goals.
- Status quo bias: Do nothing rather than make a change.
- Endowment bias: People value asset more when they hold rights to it.
- Regret-aversion bias: Avoid pain of regret associated with bad decisions.

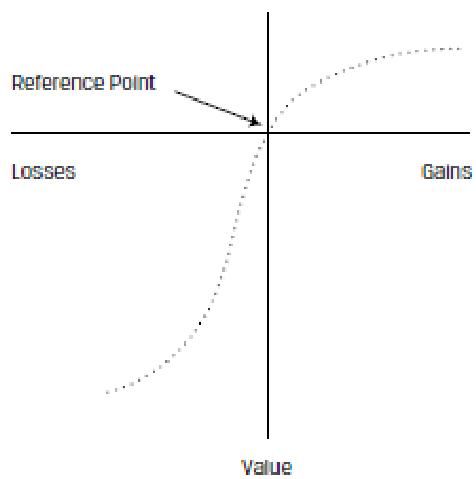
4.1 Loss-Aversion Bias

Loss-aversion bias is a bias in which people prefer avoiding losses over achieving gains. When comparing absolute values, the utility derived from a gain is much lower than the utility given up with an equivalent loss. Stated simply, people hurt more when there is a loss and are less happy when they gain the same value.

Loss-averse behavior is explained as the evaluation of gains and losses based on a reference point. A value function that passes through this reference point is seen in Exhibit 1 below. The x-axis has gains/losses, and the y-axis has value. Say we have a 5% gain, the positive value is 5 units, but for the same amount of loss, 5%, notice from the exhibit that the value is much higher, say -15 units. So, the value of gain is not symmetric to the value of the loss.

The value function is s-shaped and asymmetric; it has a greater impact of losses than gains for the same variation in absolute value. This utility function implies risk-seeking behavior in the domain of losses (below the horizontal axis) and risk avoidance in the domain of gains (above the horizontal axis).

Exhibit 1 Value Function of Loss Aversion



An important concept based on this utility function is *disposition effect*: the holding of investments that have experienced losses too long, and the selling of investments that have experienced gains too quickly. Hence the resulting portfolio may be riskier than the one based on the investor's risk/return objectives.

Consequences of loss aversion:

- Hold investments in a loss position for a longer period of time than is justified by fundamentals in the hope that they will return to breakeven.
  For example, you bought a stock for 20, and after a while, its price declined to 15. This was a reasonable price to sell, given the fundamental value of the stock. But you held on to the stock in the hope of regaining value rather than recognizing a loss. Such behavior leads to a riskier portfolio than one that was based on risk/return objectives and fundamental analysis.
- Sell investments in a gain position earlier than justified by fundamentals. Trade excessively as a result of selling winners.

Here, say you bought a stock that increased by 10%, and you sold it because you were afraid that price would decrease if you held on. As a result, you would trade excessively. Fundamentals should justify your selling decisions.

- Limit the upside potential of a portfolio by selling winners and holding losers.

Detecting and overcoming loss aversion:

- Use a disciplined approach to investment based on fundamental analysis.

4.2 Overconfidence Bias

Investors demonstrate unwarranted faith in their own intuitive reasoning, judgments, and/or cognitive abilities. For example, people generally think they do a better job of estimating probabilities than they actually do. This is known as illusion of knowledge bias.

The two basic types of overconfidence bias within the illusion of knowledge bias are *prediction overconfidence* and *certainty overconfidence*.

- Prediction overconfidence happens when the confidence intervals assigned (for investment predictions) are too narrow. For example, when estimating a stock's value, you are overconfident that there will be too little variation—use a narrower expected payoffs' range and a lower standard deviation of returns than justified by the fundamental value.
- Certainty overconfidence occurs when the probabilities for outcomes are too high because FMPs are too certain of their judgments. For example, if you are certain that the investment will increase in value, the probability that you will assign to this scenario will be too high. Thus, you do not consider the prospect of a loss and predict high returns with virtual certainty.

Self-attribution bias is a bias in which people take credit for successes and blame exogenous factors (such as bad luck) for failures. This bias can be divided into two subsidiary biases: *self-enhancing* and *self-protecting*. Self-enhancing bias describes people's propensity to claim too much credit for their successes. Self-protecting bias describes the denial of responsibility for failures.

Consequences of overconfidence bias:

- Underestimate risks and overestimate expected returns.
- Hold poorly diversified portfolios.

Detecting and overcoming overconfidence bias:

- Review trading records, identify the winners and losers. Look at your data, the stocks you bought, how they performed, and if it was in line with your expectations. Keeping a track of your performance history can adjust the overconfidence bias.
- Calculate portfolio performance over at least two years.

- There is an old Wall Street adage, "Don't confuse brains with a bull market." In a bull market, an average person will select stocks that will do well. Because in a bull run of the market, all stocks will be going up, and it does not take a genius to pick winning stocks.
- Conduct a post-investment analysis of both successful and unsuccessful investments.

4.3 Self-Control Bias

Self-control bias is a bias in which people fail to act in their long-term best interest because of a lack of self-discipline. Individuals are impulsive, and there is an inner conflict between short-term satisfaction and achievement of some long-term goals. For example, "People pursuing the CFA charter may fail to study sufficiently because of short-term competing demands on their time."

Consequences of self-control bias:

- Save insufficiently for the future. If an individual does not have self-control, he may spend on, say, a new car or a vacation, instead of saving for retirement.
- Upon realizing that the savings are insufficient, FMPs may accept excessive risk in their portfolios to generate higher returns. This is done to compensate for inadequate savings, which puts the capital base at risk.
- Borrow excessively to finance current consumption.

Detecting and overcoming self-control bias:

- Create a proper investment plan.
- Execute the plan. Adhere to a saving plan and an appropriate asset allocation strategy.

4.4 Status Quo Bias

Status quo is an emotional bias in which people do nothing (i.e., maintain the "status quo") instead of making a change. People are generally lazy and keep things the same. They do not necessarily look for opportunities where a change is beneficial. A phrase commonly used can be understood in the context of status quo, "if it ain't broke, don't fix it." This saying means leave something, avoid correcting or improving it.

Status quo bias is often discussed in tandem with endowment and regret-aversion biases (described later) because the outcome of the biases may be similar. However, the reasons differ among the biases. In the status quo bias, positions are maintained primarily due to inertia rather than conscious choice. In the endowment and regret-aversion biases, the positions are maintained because of conscious, but incorrect choices.

Consequences of status quo bias:

- FMPs unknowingly maintain portfolios with risk characteristics that are inappropriate for their circumstances.

- They fail to explore other opportunities.

Detecting and overcoming status quo bias:

- Education is crucial.
- Quantify the risk-reducing and return-enhancing advantages of diversification and proper asset allocation.

4.5 Endowment Bias

Endowment bias is an emotional bias in which people value an asset more when they hold rights to it than when they do not. For example, a person owns his house, and it has a specific value to him. After some time, he decides to buy another house next door, very similar to his house. But he feels the price quoted for it is too high. However, if he were to sell his own house, he would perhaps ask for a higher price. Standard economic theory states that similar assets should be bought/sold at the same price. Because he has lived and owned this house for the past 30 years, endowment bias has affected his attitudes toward it. Effectively, ownership "endows" the asset with added value.

Endowment bias may also occur when someone inherits or purchases securities. The value in their minds is more than what it should be. Consequently, FMPs may irrationally hold on to securities or assets they already own even though it violates the law of one price.

Endowment bias can be combined with the status quo bias. Clients are reluctant to sell securities bequeathed by previous generations.

Consequences of endowment bias:

- Fail to sell off certain assets and replace them with other assets.
- Maintain an inappropriate asset allocation.
- Continue to hold assets with which they are familiar.

Detecting and overcoming endowment bias:

- In the case of inherited investments, ask such a question, "If I were given an equivalent amount in cash, how would I invest it now?"
- Address emotional attachment.

4.6 Regret-Aversion Bias

Regret-aversion bias is a bias in which people tend to avoid making decisions out of fear that the decision will turn out poorly. For example, many years ago, there was a saying that no one gets fired for selecting IBM. Because IBM was a big company, and it was considered safe to buy merchandise from it. Even if your decision went wrong, you could save face by saying, "I bought from the biggest company," which is commonly accepted as the appropriate thing to do. Generally, people try to avoid the pain of regret associated with bad decisions. Thus, no action becomes the default decision.

Consequences of regret-aversion bias:

- Be too conservative in their investment choices because of poor outcomes on risky investments in the past. To avoid the regret of coping with another bad decision, FMPs will invest in low-risk assets failing to reach investment goals in the long-term.
- Engage in herding behavior. FMPs may feel safer investing in popular investments. It seems safe to be with the crowd; if the investment turns out to be unsuccessful, everyone faces the loss, reducing emotional pain.

Detecting and overcoming regret-aversion bias:

- Education is crucial.
- Quantify the risk-reducing and return-enhancing advantages of diversification and proper asset allocation.
- Recognize that losses happen to everyone; keep in mind long-term objectives.

5. Behavioral Finance and Market Behavior

Traditional finance assumes that markets are efficient, that they instantly and fully incorporate all information into asset prices. However some persistent market patterns such as momentum, value, bubbles, and crashes challenge this assumption. In this section we will use behavioral finance to understand why these patterns occur.

5.1 Defining Market Anomalies

Market anomalies are persistent deviations from the efficient market hypothesis (EMH). Anomalies are identified by persistent abnormal returns that differ from zero and are predictable in direction. Possible explanations of anomalies are:

- Shortcomings in the underlying asset pricing model – For example, the CAPM model uses return relative to risk (beta) incurred. If the mispricing of security is due to other types of risks not considered in the model, then the market is not anomalous; the model is limited. Examples are low returns following initial public offerings (called the "IPO puzzle") and the positive abnormal returns in the 12 months after a stock split.
- Small sample involved.
- Statistical bias in selection or survivorship, or data mining that overanalyzes data for patterns and treats spurious correlations as relevant.
- Temporary disequilibrium behavior - unusual features that may survive for years but ultimately disappear.

Despite the challenges, behavioral finance has identified several market anomalies and offered explanations based on individual investors' behavioral biases. Three such anomalies are momentum, bubbles, and crashes, and value stocks vs. growth stocks.

5.2 Momentum

Momentum (or trending) occurs when future price movements are correlated with recent past prices. Correlation is positive in the short term, up to two years, but for longer periods of two to five years, correlation is negative, and returns revert to the mean.

Why does this phenomenon occur?

Momentum can be partly explained by availability and hindsight biases.

Several studies have found that traders have faulty learning models in which their reasoning is based on their most recent experience. This is also referred to as availability bias. The *recency effect* is the tendency to recall recent events more vividly and give them undue weight. If the price of a security rises for a period, investors extrapolate this rise to the future. Recency bias causes investors to place too much emphasis on small samples because of the readily available information.

Regret is the feeling that an opportunity has been missed and is related to *hindsight bias*. Hindsight bias is the tendency to see past events as having been predictable. Suppose a mutual fund or stock did well a year ago; the regret of not owning this well performing asset can drive investors to remedy it by purchasing the asset. These behavioral factors can explain short-term year-on-year trending and overtrading creating a *trend-chasing effect*. Investors have a bias to buy investments they wish they had owned the previous year.

5.3 Bubbles and Crashes

Bubbles and crashes are defined as periods of unusual positive or negative asset returns. Bubbles and crashes appear to be panics of buying and selling. A continuous increase in asset price is fueled by further expectations of increase; asset prices become decoupled from economic fundamentals. Bubbles typically develop more slowly relative to crashes, which can be rapid. This asymmetry is because of the behavioral factors involved. A crash is a fall of 30% or more in asset prices over several months. Recent examples are the technology bubble of 1999–2000 and the residential property boom of 2005–2007.

Rational and behavioral finance explanations for asset bubbles is given in Example 16. Both managers did not understand the risks involved in the technology stock bubble and exhibited the illusion of control bias.

Example: Investor Behaviour in Bubbles
(*This is Example 16 in the curriculum.*)

Consider the differing behavior of two managers of major hedge funds during the technology stock bubble of 1998–2000:

The manager of Hedge Fund A was asked why he did not get out of internet stocks earlier even though he knew by December 1999 that technology stocks were overvalued. He replied, "We thought it was the eighth inning, and it was the ninth. I did not think the NASDAQ composite would go down 33% in 15 days." Faced with losses, and despite his previous strong 12-year record, he resigned as Hedge Fund A's manager in April 2000.

The manager of Hedge Fund B refused to invest in technology stocks in 1998 and 1999 because he thought they were overvalued. After strong performance over 17 years, Hedge Fund B was dissolved in 2000 because its returns could not keep up with the returns generated by technology stocks.

The behavioral biases associated with bubbles and crashes are

- Overconfidence is, almost by definition, abundant during bubbles (but not necessarily in crashes). This bias causes investors to overestimate expected returns, underestimate risk, trade excessively, and hold undiversified portfolios. The overconfidence and excessive trading are linked to *confirmation bias* and *self-attribution bias*. Confirmation bias exists because investors can select news that supports an existing decision or investment.
- Even if winners are being sold too soon in a bubble, sales will be profitable because of a rising market. Investors can demonstrate hindsight, in which individuals can reconstruct prior beliefs and deceive themselves that they are correct more often than they truly are.
- Regret aversion is also present in a bubble, as investors may think they are "missing out" on profit opportunities.
- As a bubble unwinds, there can be underreaction that can be caused by anchoring when investors do not update their beliefs sufficiently.
- There can be cognitive dissonance, ignoring losses, and attempting to rationalize flawed decisions as a bubble unwinds. Investors may initially be unwilling to accept losses.
- In crashes, the disposition effect leads investors to hold on to losers and postpone regret. This leads to an initial underreaction to bad news but later causes capitulation and acceleration of share price decline, which results in a crash.

5.4 Value

Studies show that value stocks outperform growth stocks over long periods. Value stocks are typically characterized by low price-to-earnings ratios, high book-to-market equity, and low price-to-dividend ratios. A growth stock has the opposite characteristics of a value stock. For example, growth stocks are characterized by low book-to-market equity, high price-to-earnings ratios, and high price-to-dividend ratios.

The Fama and French three-factor model incorporates additional factors, size, and value, alongside market beta in their asset pricing model to explain this anomaly and other apparent anomalies. Fama and French claim the value stock anomaly disappears in their three-factor model. They believe that the factors are associated with risk exposures that show the greater potential to suffer distress during economic downturns. In other words, it is not the mispricing of value and growth stocks that lead to the difference in their performance but the difference in risk between the two classes of stocks.

Some studies present anomalies as mispricing rather than risk. These studies identify the emotional factors involved in valuing stocks. The **halo effect**, for example, is based on a favorable evaluation of some characteristics relative to others. This view is related to representativeness that can lead investors to extrapolate recent past performance into future returns. A company with a sound growth track and share price performance might be seen as a good investment with higher returns than its risk would merit. This means that the risk of growth stocks is underestimated.

Emotions play a role in estimating risk and expected return of stocks. An emotional rating in a company leads investors to perceive the company's stock as less risky. Although the capital asset pricing model assumes risk and expected return are positively correlated, many investors behave as if the correlation is negative, expecting higher returns with lower risk.

The emotional attraction of a stock can be enhanced by personal experience of products, the brand's value, and the proximity of its head offices to the analyst or investor. This last issue is known as the **home bias** anomaly, whereby domestic securities are favored over international securities in global portfolios.

Summary

LO: Compare and contrast cognitive errors and emotional biases.

- Cognitive errors can be broken down into two types
    - Belief perseverance biases: People hold on to original beliefs, and they react selectively to new information. Examples: Conservatism, Confirmation, Hindsight, Illusion of control, Representativeness.
    - Information processing biases: Also called statistical errors. These errors can occur because people process information incorrectly or because of memory errors or faulty reasoning. Examples: Framing, Anchoring & Adjustment, Mental Accounting, Availability.
- Emotional biases are influenced by feelings and emotion and usually related to human behavior to avoid pain and produce pleasure; arise spontaneously due to attitudes and feelings; are less easy to correct and can only be "adapted to." Examples: loss-aversion, overconfidence, self-control, endowment, regret aversion, and status quo.

LO: Discuss commonly recognized behavioral biases and their implications for financial decision making.

Belief perseverance Bias	Description	Examples/Implications
Conservatism	Maintain prior views by inadequately incorporating new information	• Hold winners or losers too long • Under-react to new information • exhibit discomfort or difficulty in processing new information
Confirmation	Look for and notice what confirms prior beliefs	• Focus on confirmatory/positive information about existing investments • Over-react to confirmatory/positive information • Hold under-diversified portfolio
Hindsight	See past events as having been predictable	• Overestimate the degree to which a prior event was predictable
Illusion of Control	False belief that we can influence or control outcomes	• Feeling of control over company where one works • Hold under-diversified portfolio
Representativeness	Classify new information based on past experiences	• Look for patterns in new information • Over-optimism about a past winner • Treat small sample as "representative" of entire population

		• Invest in companies that remind one of successful clients • Over-react to new information and neglect base rate • Excessive trading and high manager turnover (owing to focus on short-term performance)

Information-Processing Biases	**Description**	**Examples/Implications**
Framing	Answer question differently based on how it is asked/framed	• Exhibit risk-averse (risk-seeking or loss-aversion) attitude when outcomes are framed in terms of gains (losses)
Anchoring and Adjustment	developing estimates based on "anchor" value (e.g., target price) and adjusting decisions up or down based on that value	• Place high weight on anchor • Under-react to new information • Influenced by purchase price or arbitrary price levels
Mental Accounting	Treat one sum of money different from another depending on source or use	• Investing some money very conservatively and the rest in speculative stocks. • Ignore correlations among various assets and total return • Hold suboptimal portfolio due to inefficient asset allocation
Availability	Influenced by how easily outcome comes to mind	• Place high weight on easily available information → influenced by advertising • Select alternatives with which one has greater resonance; select alternatives that are easily retrievable • Focus on a limited set of investments ("categorization") • Make investment decisions based on their familiarity with the industry or country ("narrow range of experience")

Emotional Bias	**Description**	**Examples/Implications**
Loss Aversion	Prefer avoiding losses over achieving gains	• Hold on to losing stocks too long and sell winning stocks too early (also called "disposition effect")

Overconfidence	Unwarranted faith in one's abilities (Illusion of knowledge; self-attribution)	• Excessive trading • Narrow confidence intervals • Assign high probability of success
Self-Control	Fail to act in pursuit of long- term goals	• Focus on short-term satisfaction • Fail to save enough for the future
Endowment	Exhibit an emotional attached to the asset owned	• Shares in father's company a source of family pride • People value asset more when they hold rights to it • Hold inherited/purchased securities
Regret Aversion	Avoid pain of regret associated with bad decisions	• Hold losing positions for too long • Prefer low risk assets • Engage in "herding behavior" • Prefer maintaining positions in familiar investments
Status Quo	Do nothing rather than make a change	• Hold on to securities even if they are inconsistent with risk/return objectives; trade very infrequently

LO: Describe how behavioral biases of investors can lead to market characteristics that may not be explained by traditional finance.

Observed Market Behavior	**Behavioral Explanation**
Momentum or trending effect	Herding behavior Availability bias: more recent events easily recalled and given relatively high weight (recency effect) Hindsight bias → regret → trend-chasing effect
Bubbles	Overconfidence bias (illusion of knowledge and self-attribution) leads to underestimation of risk and over-trading
Crashes	Disposition effect in the context of loss aversion bias: tendency to sell winners quickly and hold on to losers too long
Value stocks outperform growth stocks in the long-run	**Halo effect:** tendency of people to generalize positive views/beliefs about one characteristic of a product/person to another characteristic; related to ***representativeness bias*** refers to classifying new information based on past experiences. Underestimating risk and overestimating returns based on emotional factors.

LM06 Introduction to Risk Management

Version 1.0

1. Introduction

Investors often assume higher risk in the pursuit of higher returns. Businesses and investors manage risk, whether consciously or not, in every investment decision they make. This reading lays the fundamentals of risk management from the perspective of both businesses and individuals.

Some of the important concepts addressed in this reading include:

- What is risk management and why is it important?
- How businesses and individuals manage risk?
- The principles behind both enterprise and portfolio risk management.
- How an entity's goals are affected by risk and how risk management decisions produce better results?
- Identifying the various risks and the tools used by an organization to manage risk.

2. Risk Management Process

Risk is the exposure to uncertainty. Risk driver is the underlying risk. Risk position is the description or quantification of the risky action taken. Risk exposure is the extent to which an entity is sensitive to underlying risks. In other words, risk exposure is the risk position multiplied by the risk driver.

Risk management is the process by which an organization or individual defines the level of risk to be taken (risk tolerance), measures the level of risk being taken (risk exposure), and adjusts the latter toward the former, with the goal of maximizing the company's or portfolio's value or the individual's overall satisfaction or utility. Ideally, risk exposure should roughly be equal to risk tolerance. Risk management is **not** about minimizing risk, but about actively managing risks to achieve goals. The focus is on risk management (as opposed to return management) because it is possible to manage risk, but it is not always possible to manage returns.

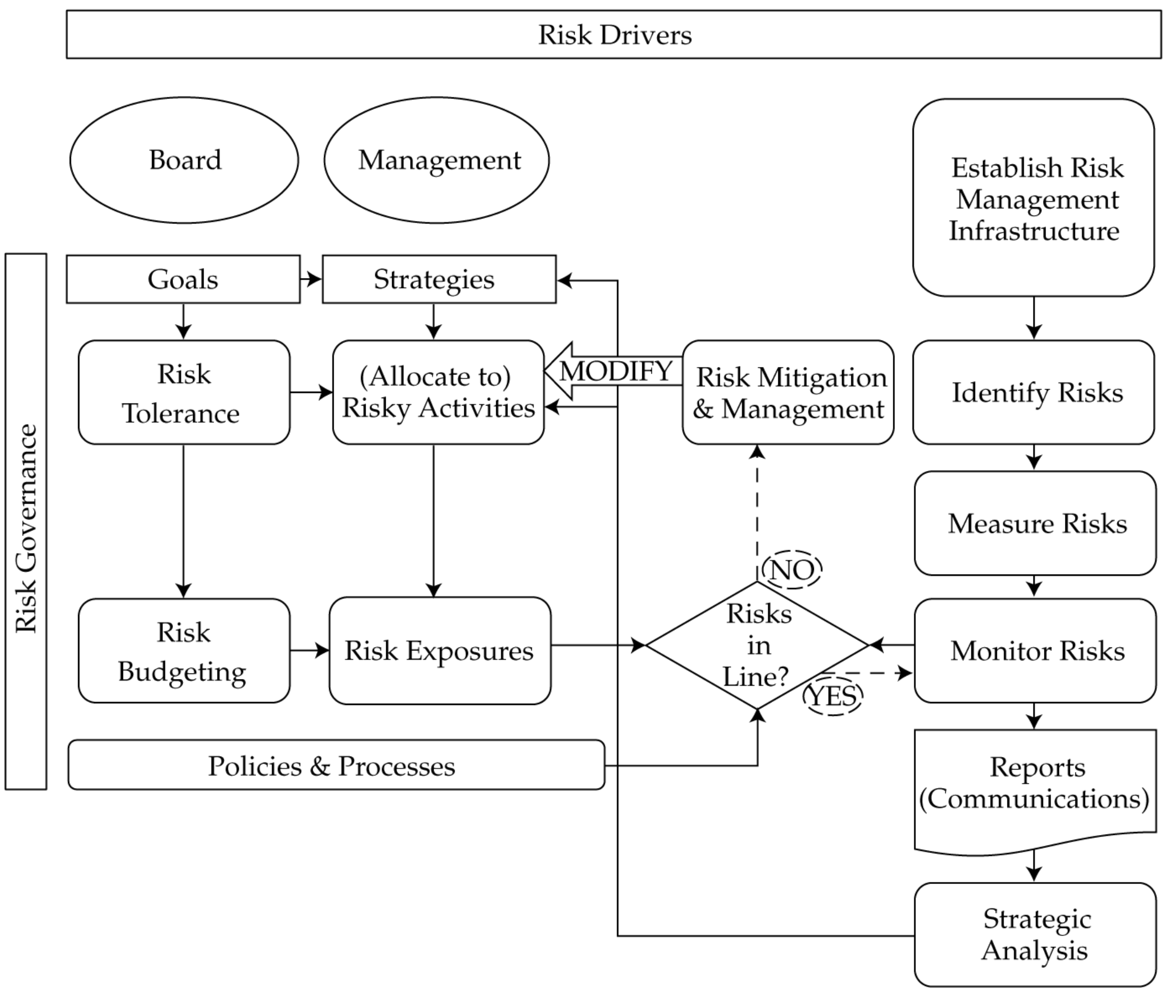
3. Risk Management Framework

A risk management framework is the infrastructure, processes, and analytics needed to support effective risk management in an organization. Any risk management framework should include the following factors:

- Risk governance: This top-down process lays the foundation for risk management in an organization. Good governance ensures that the risk tolerance level is set for an organization and provides risk oversight.
- Risk identification and measurement: This is the quantitative and qualitative assessment of all sources of risk to an organization.
- Risk infrastructure: This refers to the people and the systems required to track risk exposures and to perform risk analysis.

- Defined policies and processes: These are limits, requirements, constraints, and guidelines to ensure that an organization's risky activities are within its risk tolerance levels.
- Risk monitoring, mitigation, and management: This primarily involves identifying, measuring and continuously monitoring risk exposure of an organization. If risk exposure is not aligned with pre-defined risk tolerance, then necessary action is taken to restore balance between the two.
- Communications: Critical risk issues must be continually communicated across all levels of an organization. Risk tolerances must be communicated to managers. Risk metrics must be reported in a timely, easy-to-understand manner. A feedback loop with the governance body should be present to ensure that risk guidance is validated and communicated to the rest of the organization.
- Strategic analysis or integration: The objective of this analysis is to use risk management to increase the overall value of the business.

The diagram below shows the risk management framework in an enterprise context.



4. Risk Governance: An Enterprise View

4.1 An Enterprise View of Risk Governance

Risk governance is a top-down process that defines risk tolerance and provides guidance to align risk with enterprise goals. It includes guidance on unacceptable risks and worst losses that can be tolerated.

An enterprise risk management perspective deals with the whole organization. The governing body drives the risk framework in the following ways:

- It determines the goals of the organization.
- It is responsible for providing risk oversight to ensure that value is maximized.
- It determines the risk tolerance level; which risks are acceptable, which risks to mitigate, and which risks are unacceptable. The process includes guidance on worst losses that can be tolerated for every scenario.

Elements of good risk governance are as follows:

- To provide a forum where management can discuss the risk framework and key issues that come up during execution.
- Form a risk governance committee to oversee the implementation of the framework at an operational level relative to the high level oversight by the governance body.
- Appoint a chief risk officer (CRO) to build and implement the risk framework for the entire enterprise.

5. Risk Tolerance

Risk tolerance identifies the extent of losses an organization is willing to experience. Risk tolerance focuses on the appetite for risk of an organization in its pursuit of achieving goals and maximizing value. The process involves defining:

- Which risks are acceptable and which risks are not acceptable?
- How much risk can the entity be exposed to?

The risk tolerance decision begins with two different analyses:

- Inside view: What shortfalls within the organization will cause it to fail or not achieve certain goals?
- Outside view: What uncertain forces is the organization exposed to?

Using these two views in conjunction, the board will:

- Define which risks to take and which risks not to take.
- Determine the risk appetite: how much of these risks to take.
- Communicate risk tolerance before a crisis.
- Provide a high-level guidance to management in strategizing and choosing activities.

There are no standard formulas to determine the risk tolerance of a company. Some of the factors that will help a board determine its risk appetite are as follows:

- Company’s areas of expertise and goals.

- Ability to respond dynamically to adverse events: The higher the ability, the higher the level of risk tolerance.
- The amount of loss a company can bear without impacting its status as a going concern.
- The company's position in the industry, and how it fares relative to its competitors.
- Government and regulatory landscape where the company operates.
- Quantitative analyses such as scenario analyses, macro analyses. etc.

Once risk tolerance is determined, the objective of the overall risk framework should be to align risk exposure with the enterprise's risk appetite.

6. Risk Budgeting

Risk budgeting helps determine how or where risks are taken and quantifies tolerable risks by specific metrics; risk budgeting should drive hedging strategies (not the other way round).

Risk budgeting allocates investments or assets by their risk characteristics rather than by a common classification of asset class such as stocks, bonds, real estate, etc. For example, the risk view of a portfolio might be that it is driven 70% by global equity returns, 20% by domestic equity returns, and 10% by interest rates, or a portfolio that has 45% illiquid and 55% liquid securities.

<u>How risk budget is measured</u>

- It can be a complex, multi-dimensional measure that evaluates risks based on their asset classes such as equity, commodities, and real estate and then allocates investment by their asset class.
- It can also be a simple, one-dimensional risk measure such as standard deviation, beta, and value at risk and scenario loss.
- Risk factor approaches are also used, in which exposure to various factors is used to determine risk premiums.
- Example: portfolio beta is limited to 1.

One of the biggest benefits of the risk budgeting process is that it forces a firm to consider risk trade-offs. By adopting risk budgeting, it helps a business to:

- Choose the project with the highest return per unit of risk.
- Choose between doing less risky investments and more risky investments whose risks have been hedged.
- Compare active versus passive strategies. This helps businesses make decisions to add active value while staying within the risk tolerance levels.

7. Financial vs. Non-Financial Risk

There are two categories of risks: financial risks and non-financial risks.

7.1 Financial Risks

Financial risks are the risks that originate from financial markets, such as changes in interest rates or prices. There are three primary types of financial risks:

- **Market risk**: This risk arises from movements in stock prices, exchange rates, interest rates, and commodity prices. Any changes in fundamental economic conditions, events in the industry or economy, give rise to market risk.
- **Credit risk**: This is the risk that a counterparty will not pay an amount owed on an obligation, such as a bond, loan, derivative, to another party.
- **Liquidity risk**: This is the risk that, as a result of degradation in market conditions or the lack of market participants, one will be unable to sell an asset without lowering the price to less than the fundamental value. Liquidity risk is also known as transaction cost risk. Liquidity risk arises when the market for a specific asset becomes less liquid or the size of a position increases.

7.2 Non-Financial Risks

Non-financial risks are risks that arise from sources outside the financial markets, "from actions within the organization or from external origins, such as the environment as well as from the relationship between the organization and counterparties, regulators, governments, suppliers, and customers".. These risks also have a monetary impact on the organization. The various types of non-financial risks are discussed below:

- **Operational risk**: This risk arises from within the operations of an organization and includes both human and system or process errors. All the internal risks in an organization are collectively called operational risk. Examples of operational risk include inadequate or failed employees, programming errors, systems, and internal policies, procedures and processes making an organization susceptible to hackers, rogue trader in a brokerage firm, natural disasters that interrupt operations, and terrorist attacks. Operational risk can also arise due to extreme weather and natural disasters such as floods, earthquakes, or hurricanes.
- **Solvency risk**: This risk arises when the entity does not survive or succeed because it runs out of cash to meet its financial obligations. One example of solvency risk is what happened to Lehman Brothers in 2008 because of taking on excessive leverage.
- **Settlement risk**: This is the default risk that occurs just before payments are to be settled.
- **Legal risk**: Any risk related to the law is a legal risk. For instance, an entity may be sued by another over a transaction, or what it does or does not do, as per the contract.
- **Regulatory, accounting, and tax risk**: The three risks are collectively known as compliance risk. This risk arises when an entity fails to comply with laws, regulations, and policies set by the government or regulatory authorities.
- **Model risk**: This is the risk of a valuation error that arises from improperly using a model. For instance, using the DDM (dividend discount model) to value a company

whose growth is not constant.
- **Tail risk**: This risk arises when there are more events in the tail of the distribution.
- **Sovereign or political risk**: This risk arises when political actions negatively impact a company.

Individuals face many of the same organizational risks outlined here, in addition they also face health risk, mortality or longevity risk, and property and casualty risk.

8. Interactions between Risks

Risks are not independent of each other and there is no clear distinction between the various risks as one risk may lead to another. For example, market risk leads to credit risk, which in turn leads to settlement risk and legal risk.

Risk interactions can be non-linear and harmful. The combined risk faced is worse than the sum of the risks of the separate components. This was seen during the 2008 crisis when many investment firms were forced to shut down because of their high leverage and insolvency. Most risk models do not take into account the interactions between risks.

9. Measuring and Modifying Risk: Drivers and Metrics

9.1 Drivers

Basic drivers of risk arise from:
- Global macroeconomics: The economic policies adopted by foreign governments and central banks have a significant impact on domestic companies.
- Domestic macroeconomics: Economic activity in a country is affected by the taxes, regulations, laws, monetary and fiscal policy introduced by government and quasi-government agencies in a country.
- Industries: Government's policies expose industries to risk. For example, the government may exempt taxes on the textile industry to encourage growth in the sector, while it may levy additional taxes on the tobacco industry.
- Individual companies: There could be an issue that is specific to the company that you have invested in. For example, a lawsuit.

Using proper risk management, some of the risk can be managed, but not all of it. For risks that cannot be controlled, an entity must ensure that its risk exposure is aligned with its objective and risk tolerance.

9.2 Metrics

Risk exposure is often expressed in terms of quantitative measures. The basic metrics used to measure market risk are as follows:
- **Probability**: It is a measure of the relative frequency with which an outcome is expected to occur. For instance, the probability of a loss of 25% implies the likelihood of incurring loss, but it does not say how much the loss would be.

- **Standard deviation**: It is a measure of the dispersion in a probability distribution. That is, it gives us a range over which a certain percentage of outcomes are likely to occur. The underlying assumption is that the returns are normally distributed. Hence, it is not an appropriate risk measure for non-normal distributions. Standard deviation and variance are measures of total risk, that is, both unsystematic and systematic risk.
- **Beta or duration**: It is a measure of the sensitivity of a security's returns to the returns on the market portfolio. For instance, if beta is 1.5, then it implies that the stock is expected to go up by 1.5% when the market goes up by 1%. Beta is generally used for stock portfolios, while duration is used to measure the sensitivity of fixed-income portfolios to changes in interest rates.
- **Derivative measures**: Delta, gamma, vega, and rho are often used measures of derivative risk. **Delta** is the sensitivity of the derivative price to a small change in the value of the underlying asset. **Gamma** measures the sensitivity of the derivative to changes in delta. **Vega** measures the sensitivity of the derivative to changes in the volatility of the underlying. **Rho** measures the sensitivity of the derivative to changes in interest rates.
- **Value at risk or VaR**: VaR measures and quantifies the risk of loss in a portfolio over a specific time period. A VaR measure comprises three elements: an amount stated in units of currency, a time period, and a probability. Let us take an example of a bank with a portfolio value of $200 million. A VaR of $3 million at 5% for one day implies that the bank is expected to lose a *minimum* of $3 million in one day 5% of the time. Note that VaR only tells us the minimum expected loss; it does not state the maximum loss.
- **Conditional VaR or CVaR**: Conditional VaR is the weighted average of all loss outcomes in the statistical distribution that exceed the VaR loss. It is a measure of expected loss.
- **Expected loss given default**: This is equivalent to CVaR for a debt security.
- **Scenario analysis and stress testing**: Stress test measure the impact of change in a specific variable such as interest rate or exchange rate. Scenario analysis measure the impact of a change in multiple variables simultaneously. Scenario analysts evaluate what would happen to a portfolio if a set of conditions or market movements occur. For example, what would be the impact on a portfolio if the Fed increases interest rates and there is a significant decline in the value of the US dollar?

10. Risk Modification – Prevention, Avoidance and Acceptance

The objective of the risk manager in the risk modification stage is to align the actual risk with pre-defined levels of risk tolerance. Different approaches to manage and modify risk are discussed below.

Risk Prevention and Avoidance

The simplest approach to manage risk may be to avoid it altogether. But, it is not as simple as it appears. For example, consider an individual who invests all his retirement savings in cash to avoid the risk of volatility in equities. By doing so, he gives up any upside return potential that equities offer and protection against inflation. Sometimes, boards may take a strategic decision to avoid risks in certain business areas altogether after analyzing the risk-return trade-off, and rather focus on areas with a higher likelihood of adding value. In reality, it is difficult to take a calculated risk by offsetting the risk of loss with the benefit of gain.

When actual risk exceeds the acceptable level, the following approaches are used to manage risk.

Risk Acceptance: Self-Insurance and Diversification

Self-insurance is simply bearing the risk because the external means to eliminate the risk are costly. For business, self-insurance means setting aside sufficient capital to cover losses. An example of self-insurance is capital and loan loss reserves set aside by a bank.

Diversification: According to modern portfolio theory, diversification is an efficient way of mitigating risk.

11. Risk Modification: Transferring, Shifting, and How to Choose

Next, we will look at two approaches to transfer or sell the undesired risk to another party.

Risk Transfer

Risk transfer is the process of passing on a risk to another party, often in the form of an insurance policy. When a corporation buys fire insurance for its office building it pays a standard premium and in return the insurance company covers the damage if the office building catches fire. Hence through the insurance policy the risk of fire damage is transferred from the corporation to the insurance company.

Risk Shifting

Unlike risk transfer where the risk is transferred from one party to another, risk shifting refers to actions that change the distribution of risk outcomes. Risk shifting typically involves the use of derivatives. Derivatives are classified into two categories:

- Forward commitments. Examples of forward commitments are forward contracts, futures contracts, and swaps.
- Contingent claims. Examples of contingent claims are call options and put options.

How to Choose Which Method for Modifying Risk

Choosing which risk mitigation method to use is an important step in the risk management process. The risk-mitigation methods discussed above are not exclusive of each other. Often, companies use all methods. Some important points to consider how to choose a method are discussed below:

- Consider the cost and benefit of each option in light of the risk tolerance of the entity.
- Organizations should avoid the risks that provide few benefits at extremely high costs.
- Organizations with large free cash flow may self-insure and diversify to the extent possible.
- Insure when risks can be pooled effectively and when the cost of insurance is less than the expected benefit.
- Risk shifting is an appropriate choice for mitigating financial risks that exceed risk appetite.

Summary

LO: Define risk management.

Risk management is the process by which an organization or individual defines the level of risk to be taken (risk tolerance), measures the level of risk being taken (risk exposure), and modifies risk exposure in line with risk tolerance. The goal is to maximize the company's or portfolio's value or the individual's overall satisfaction or utility.

LO: Describe features of a risk management framework.

A risk management framework is the infrastructure, processes, and analytics needed to support effective risk management in an organization. The factors a risk management framework should include are risk governance; risk identification and measurement; risk infrastructure; risk policies and processes; risk monitoring, mitigation, and management; communication; and strategic analysis and integration.

LO: Define risk governance and describe elements of effective risk governance.

Risk governance is the top-level foundation for risk management. The governance body is responsible for setting risk tolerance and providing risk oversight.

The elements of effective risk governance include providing a forum where the management can discuss about the risk framework, the forming of a risk governance committee, and the appointing of a chief risk officer.

LO: Explain how risk tolerance affects risk management.

Risk tolerance identifies the extent of losses an organization is willing to experience. It defines which risks are acceptable, which risks are not acceptable, and how much risk an entity can be exposed to.

LO: Describe risk budgeting and its role in risk governance.

Risk budgeting quantifies tolerable risks by specific metrics. Risk budgeting allocates investments or assets by their risk characteristics rather than by a common classification of asset class such as stocks, bonds, real estate, etc.

LO: Identify financial and non-financial sources of risk and describe how they may interact.

Financial risks are the risks that originate from financial markets. Three types of financial risks include market risk, credit risk, and liquidity risk.

Non-financial risks are risks that arise from sources outside the financial markets. The various types of non-financial risks include operational risk, solvency risk, settlement risk, legal risk, regulatory risk, accounting and tax risk, model risk, tail risk, and sovereign or political risk.

Risks are not independent of each other and there is no clear distinction between the various risks as one risk may lead to another.

LO: Describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods.

The four factors that drive risk are global and domestic macroeconomics, industries, and individual companies.

Common measures of market risk include probability, standard deviation, beta or duration, derivative measures, value at risk, conditional value at risk, expected loss given default, and scenario analysis and stress testing.

Risk can be modified by prevention, avoidance, risk transfer, or shifting.

When actual risk exceeds the acceptable level, risk can be mitigated through self-insurance and diversification.

The best method to choose to modify risk depends on the benefits weighed against the costs after considering the overall risk profile.

